Cloud Tiering with FabricPool for Cost-Efficient Storage

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1 Introduction

Data growth is relentless, yet storage budgets are not keeping pace. Organizations continue to invest in high-performance AFF flash systems, and with the rise of AI workloads competing for those same resources, demand and prices are climbing faster than ever. Storage teams discover that 60-80% of the data sitting on those expensive SSDs has not been touched in months. Every cold gigabyte on flash storage is money wasted: capacity that could serve active AI and analytics workloads or simply not need to exist on premium media at all.

Traditional answers create new problems. Buying more shelves delays the inevitable. Moving data to public cloud introduces egress costs, sovereignty questions, and operational complexity. What storage teams really need is automatic, policy-driven tiering that works transparently with any S3-compatible target, whether that is an on-premises StorageGRID appliance, an ONTAP S3 bucket on an existing cluster, or a public cloud provider like AWS, Azure, or GCP.

FabricPool in NetApp ONTAP delivers exactly this flexibility. Cold data moves to low-cost object storage automatically, and hot data remains on flash for full performance, all without application changes or user intervention. New capabilities in recent releases go further: you can write data directly to the cloud tier without waiting for cooling, optimize sequential read performance for tiered data, and manage object store operations like mirroring and tagging for tighter lifecycle integration.

About this lab

In this lab, you work as a storage administrator managing FabricPool across two ONTAP clusters. You will first set up a cluster to use cloud tiering with an on-premises StorageGRID object store, walking through every step from identifying cold data to attaching the cloud tier. Then you will switch to a cluster with an existing FabricPool configuration and data already tiered to explore the results of the tiering process and some additional features and tools available to you, including Cloud Write, FabricPool mirroring, and cloud retrieval.

By the end, you will have hands-on confidence to deploy and manage FabricPool with any supported object store target.

The concepts you practice here apply across the entire NetApp ONTAP portfolio. Cloud Volumes ONTAP (in AWS, Azure, and Google Cloud), Amazon FSx for NetApp ONTAP, Azure NetApp Files, and Google Cloud NetApp Volumes all leverage the same FabricPool technology (sometimes under different names) to tier cold data automatically. In cloud environments, tiering is especially compelling: it lets you keep a small, fast SSD performance tier (the component of the storage architecture where hot data blocks are stored) while elastic object storage absorbs the bulk of your data at a fraction of the cost, turning cloud storage from a fixed expense into a usage-optimized model.

2 Lab activities

This lab explores the following topics:

- Set up FabricPool with StorageGRID
    - Review Inactive Data Reporting
    - Create an intercluster LIF
    - Create a StorageGRID bucket
    - Add a StorageGRID cloud tier
    - Attach the cloud tier to a local tier
- Manage Tiering
    - Analyze tiering results
    - Changing a Volume’s Tiering Policy
    - Enable Cloud Write
- Object store operations
    - Add a FabricPool mirror
    - Apply object tags
    - Retrieve tiered data

2.1 Set up FabricPool with StorageGRID

FabricPool lets you reclaim expensive SSD capacity by automatically moving cold data to a low-cost object store, without changing how applications access files. When the object store is an on-premises StorageGRID appliance, your data stays within the network boundary, you avoid cloud egress fees, and no FabricPool capacity license is required.

In the following activities, you configure FabricPool end-to-end: identify cold data candidates, build the network path, prepare a target bucket, and attach it as a cloud tier with a tiering policy. In Cloud Volumes ONTAP and FSx for ONTAP deployments, much of this setup is automated by the provider respectively, but understanding the underlying components helps you troubleshoot and optimize tiering in any environment.

Activities
Review Inactive Data Reporting
Create an intercluster LIF
Create a StorageGRID bucket
Add a StorageGRID cloud tier
Attach the cloud tier to a local tier

2.1.1 Review Inactive Data Reporting

Before configuring FabricPool, you should understand how much cold data exists on your local tiers. Inactive Data Reporting (IDR) tracks which data blocks have not been accessed within a configurable cooling period and displays the results in System Manager. This information helps you evaluate the potential savings from tiering and select appropriate tiering policies.

Cluster 2 has three data volumes (“data1”, “data2”, and “data3”) on the SSD local tier “cluster2_01_SSD_1”. These volumes contain data that has been inactive long enough to appear in the IDR heat map, making them ideal candidates for FabricPool tiering.

1. Double-click **Chrome** on the desktop of the Jumphost.

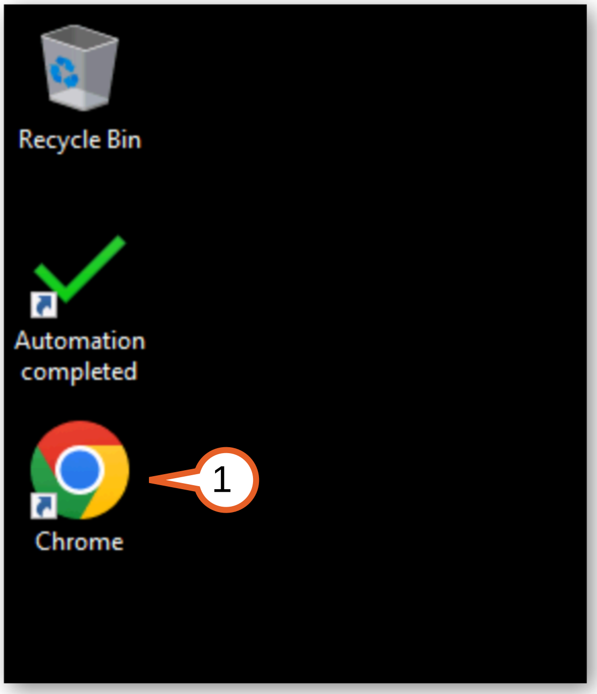

Figure 2-1:

2. From the browser toolbar, navigate to **Cluster 2**.
3. Sign into System Manager with the following credentials:
   - Username: **`admin`**
   - Password: **`Netapp1!`**

4. Select **Keep me signed in for this session.** and click **Sign in**.

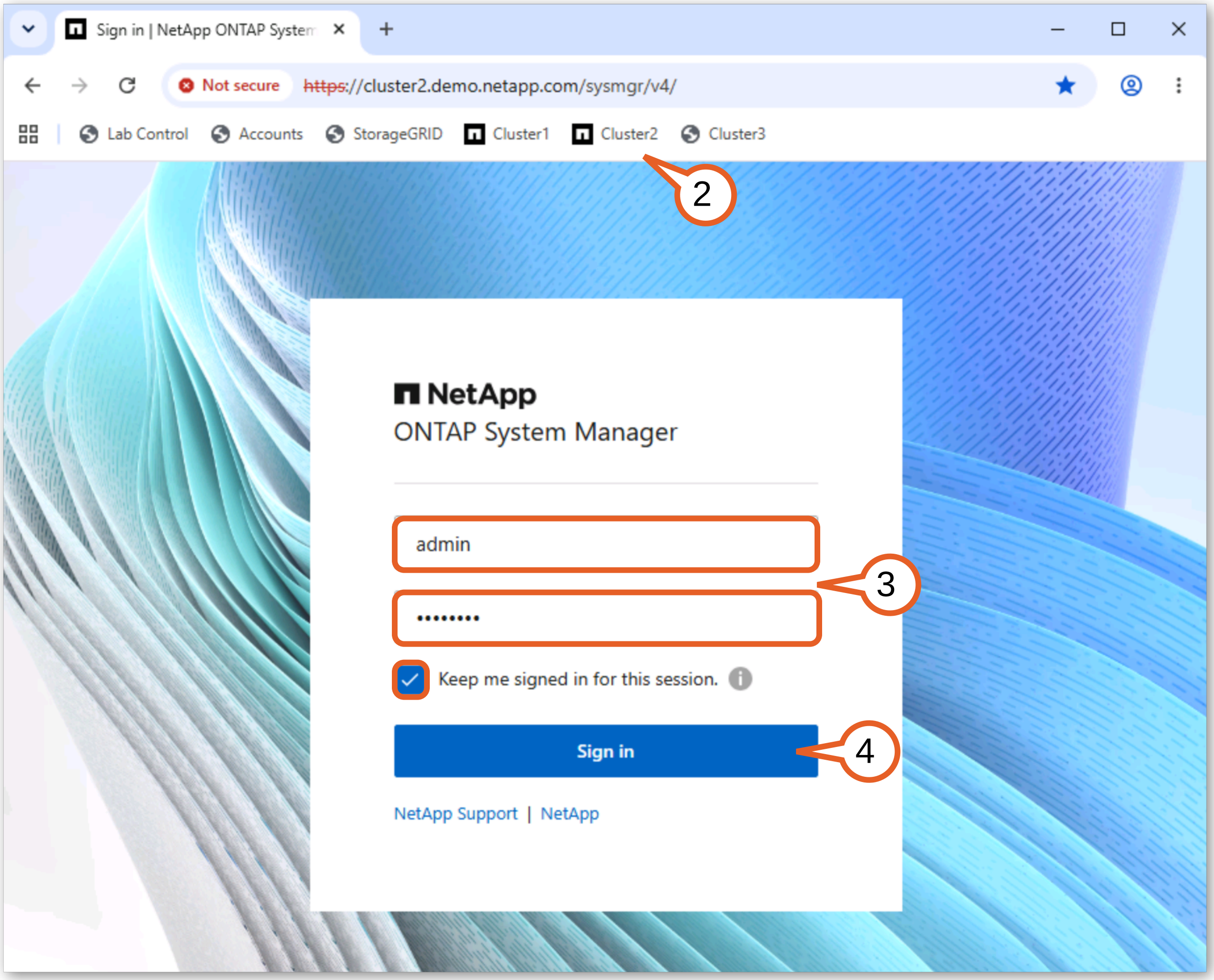

Figure 2-2:

5. In the left navigation pane, navigate to **Storage** > **Tiers**.

6. Click **cluster2_01_SSD_1** in the tiers view to open its detail page.

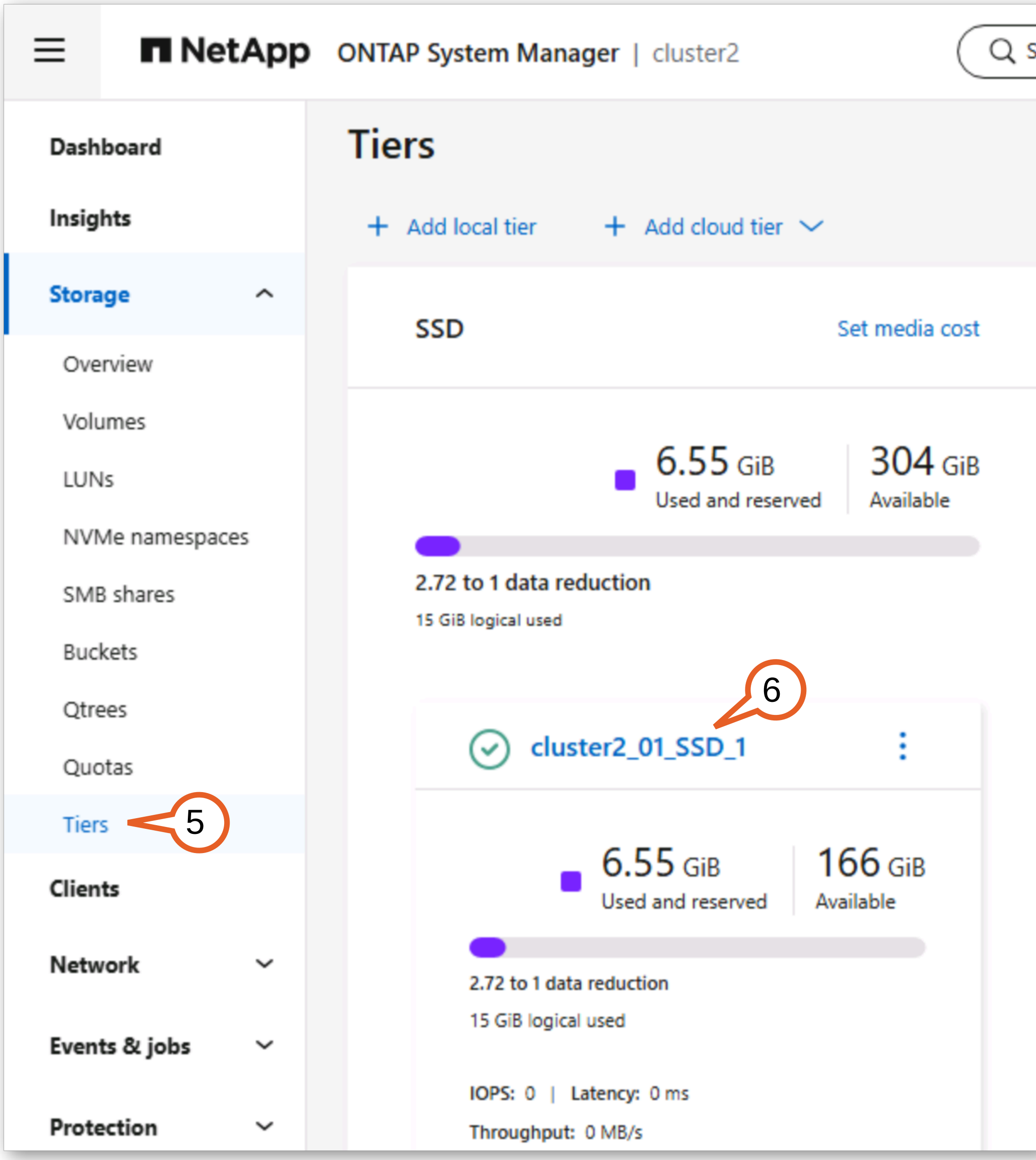

Figure 2-3:

7. Observe that a value of 7.39 GiB of inactive data is indicated on the aggregate detail page. This shows the total amount of cold data on the local tier: blocks that have not been read within the minimum cooling period. IDR runs automatically on SSD aggregates with no manual enablement required, and reports tiering potential whether or not a cloud tier is attached. Metadata blocks are never counted as inactive regardless of access patterns.

8. Click the **Volumes** tab to see per-volume inactive data.

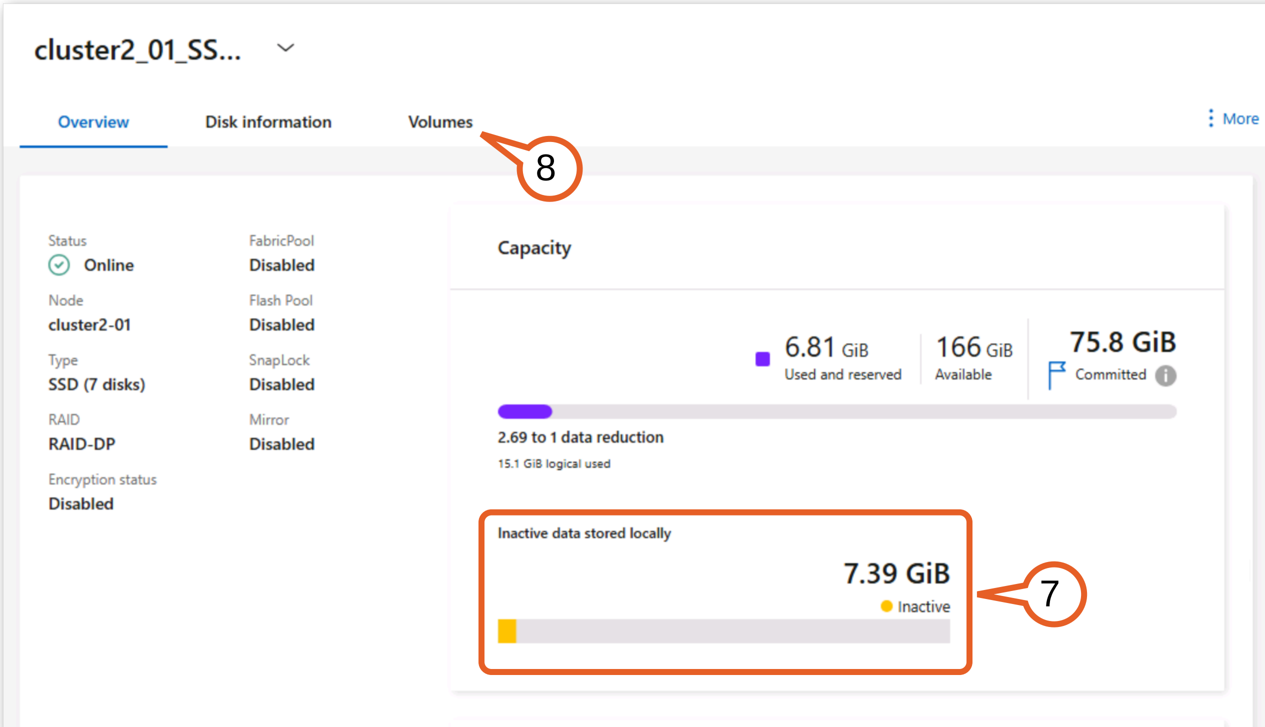

Figure 2-4:

9. Compare the inactive data values for data1, data2, and data3. Each volume shows how much of its used capacity is cold, giving you a clear picture of which volumes have the most tiering potential. data1 and data3 both have about 85% of their data as inactive, while data2 has roughly 50% inactive data. A volume with a high inactive percentage

is a strong candidate for the “auto” tiering policy, and all three of these volumes would benefit from tiering their large portion of inactive data.

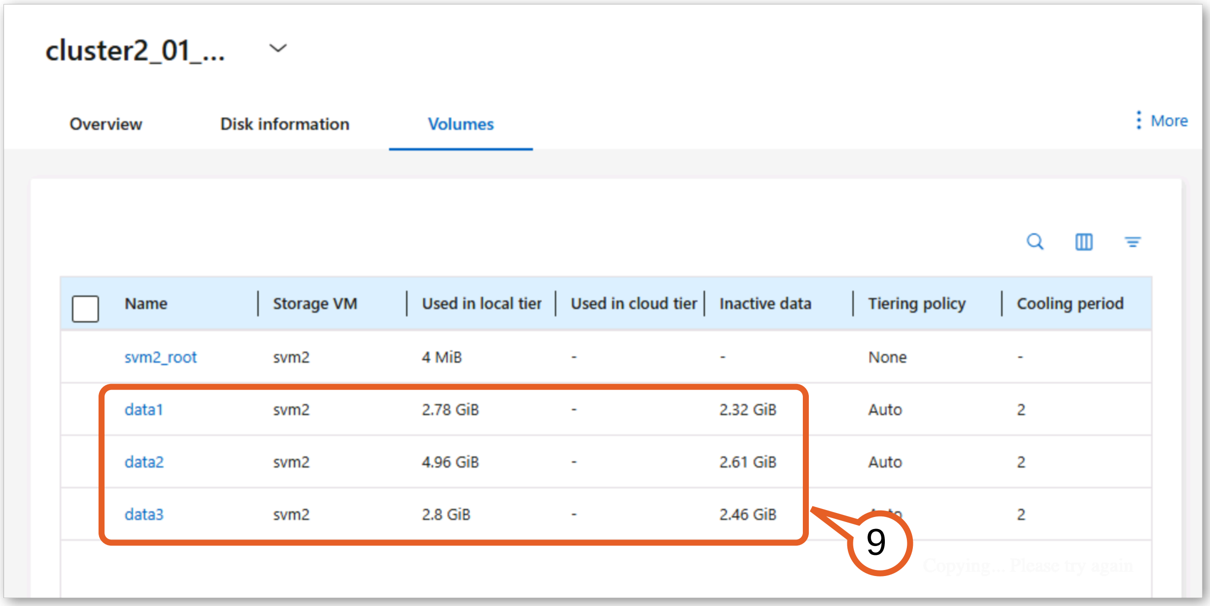

Figure 2-5:

Takeaway: IDR gives you a risk-free preview of tiering savings before you commit to any configuration changes. The per-volume breakdown helps you decide which volumes benefit most from tiering and choose appropriate policies without attaching an object store or changing any volume settings.

Note: The minimum cooling period for volumes with the auto tiering policy defaults to 31 days. In this lab environment, the cooling period has been pre-configured to two days so that tiering occurs quickly for demonstration purposes. Data must remain unaccessed for at least the configured cooling period before ONTAP considers it inactive and eligible for tiering. You can adjust this threshold per volume after attaching a cloud tier.

2.1.2 Create an intercluster LIF

FabricPool communicates with the object store over the intercluster network. Before you can add a cloud tier, each node that will tier data must have an intercluster Logical Interface (LIF) configured. This dedicated interface keeps object store traffic separate from client data access and cluster management, ensuring that tiering operations do not compete with production workloads for bandwidth.

1. In System Manager for Cluster 2, navigate to **Network** > **Overview**.

2. In the Network Interfaces section, click **+ Add** to add a new interface.

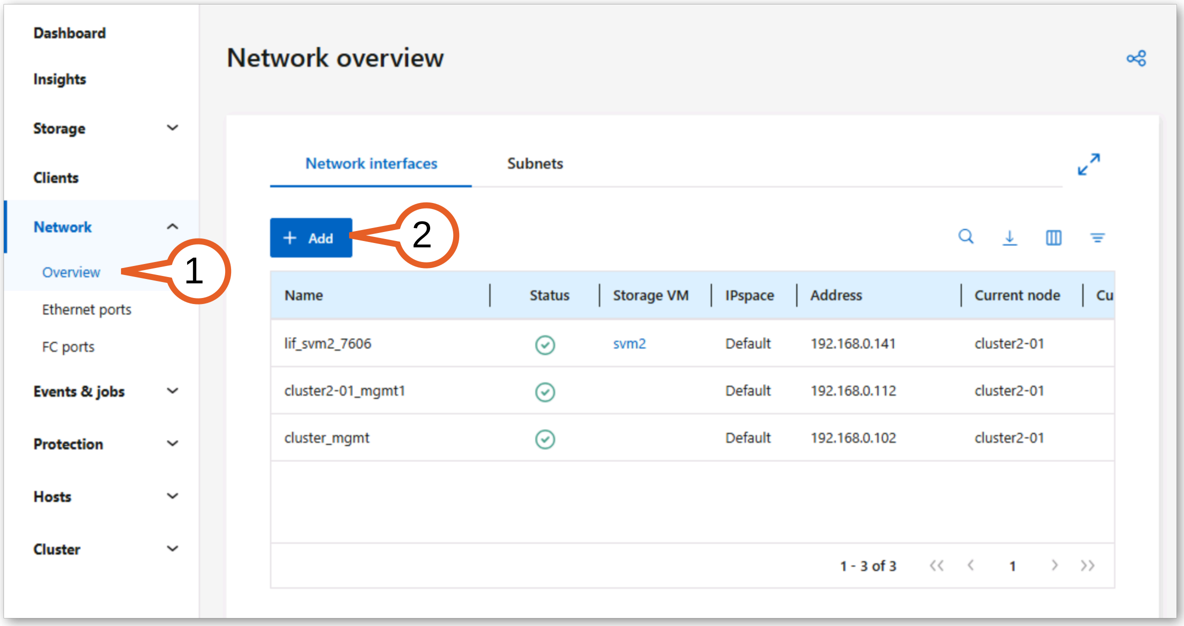

Figure 2-6:

3. Add these interface details at the top of the form:

- Interface type: **Intercluster**
- Name: **`intercluster1`**
- Home node: **cluster2-01**
- IP address: **`192.168.0.159`**
- Subnet mask: **`24`**

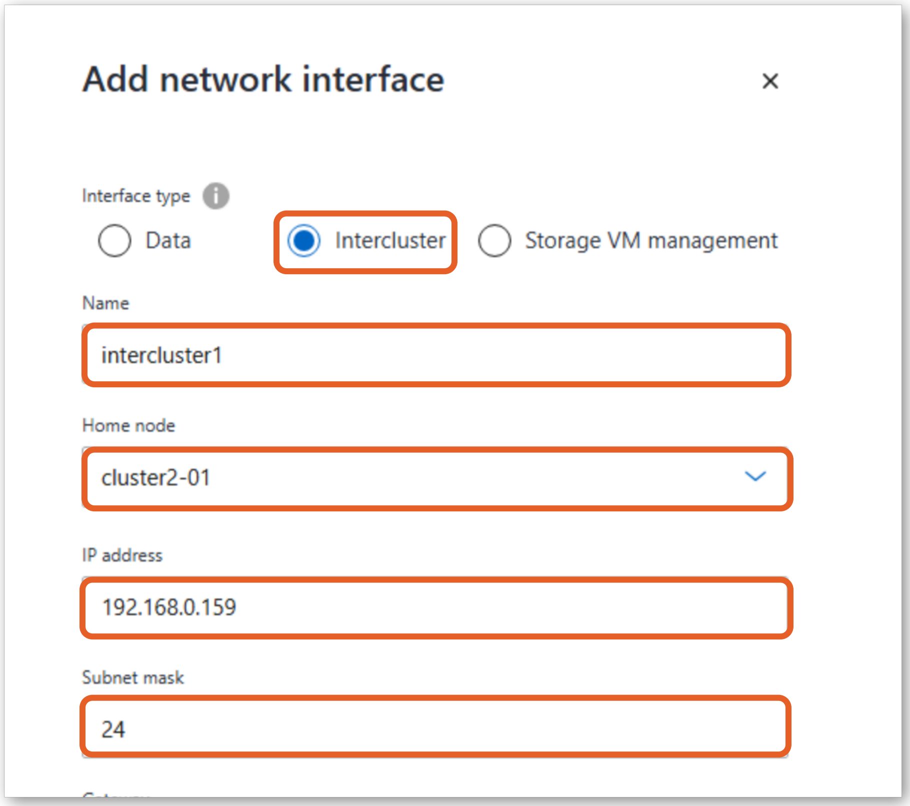

Figure 2-7:

4. Then, near the bottom, enter:

- Broadcast domain: **Default**
- Home port: **e0c**

5. Click **Save**.

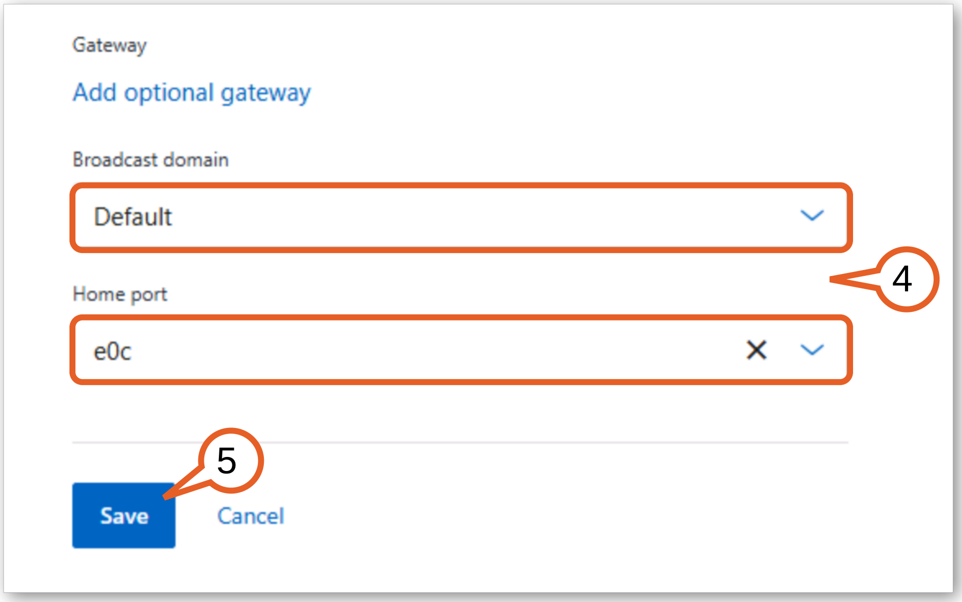

Figure 2-8:

6. Verify the new “intercluster1” LIF appears in the interface list with a status of Up, indicated by a **green check**. This confirms the intercluster network path is ready for FabricPool to communicate with the StorageGRID object store.

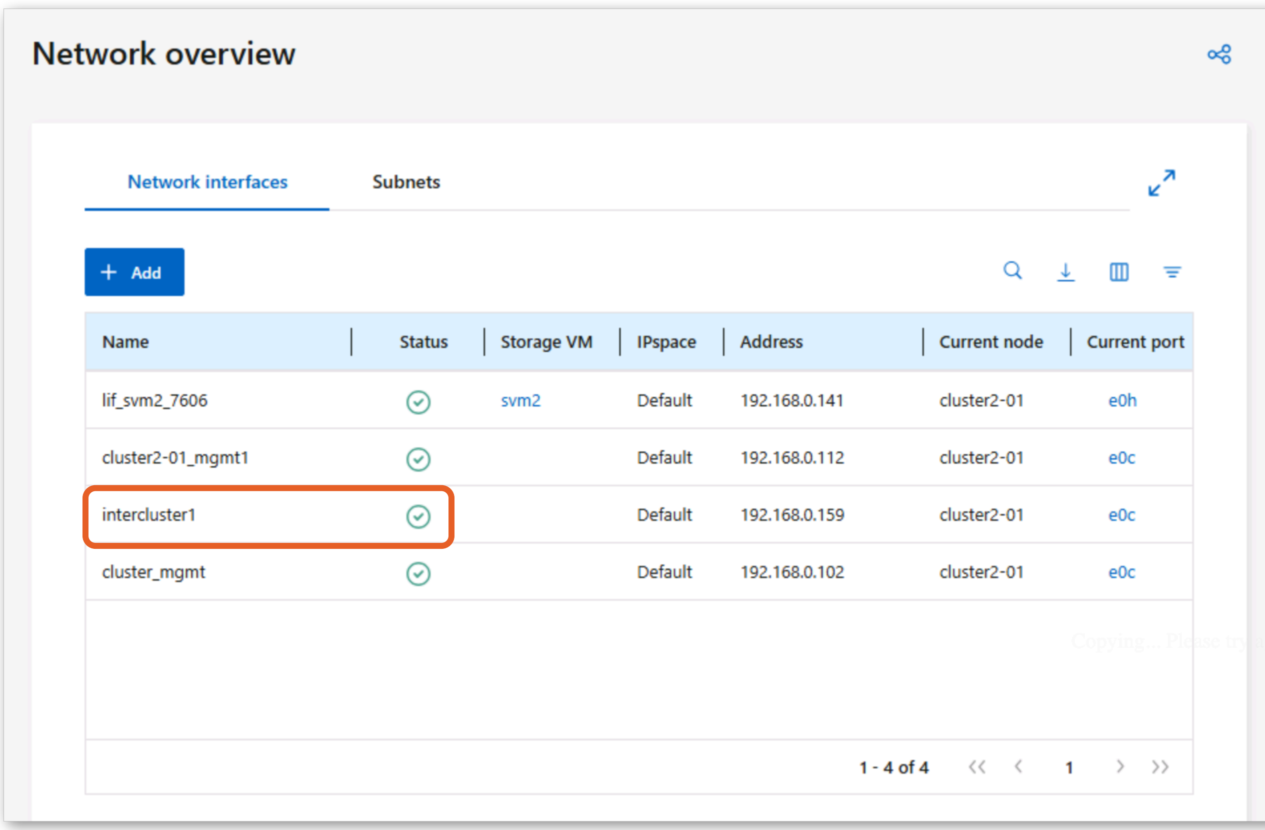

Figure 2-9:

Tip: In a multi-node cluster, you would create an intercluster LIF on every node that hosts aggregates you want to tier. This single-node lab cluster requires only one intercluster LIF.

2.1.3 Create a StorageGRID bucket

FabricPool stores tiered data in an S3 bucket on the object store. You need to create a dedicated bucket in StorageGRID for Cluster 2 to use as its cloud tier.

1. From the browser toolbar, navigate to **StorageGRID**.
2. Sign into the StorageGRID Grid Manager with the following credentials:
    - Username: **`root`**
    - Password: **`Netapp1!`**

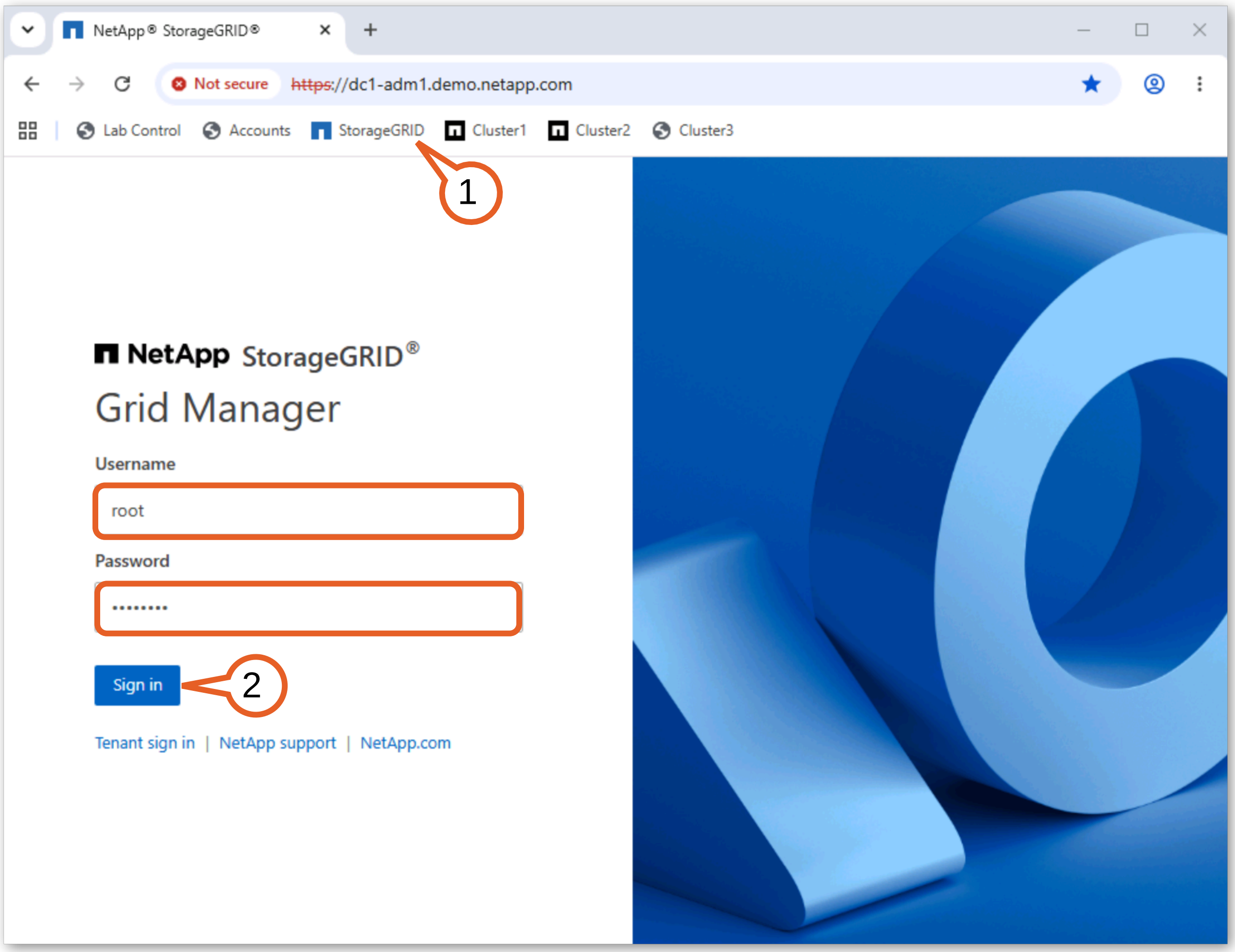

Figure 2-10:

3. In the left navigation, click **Tenants**.

4. On the Tenants page, click **LoD**.

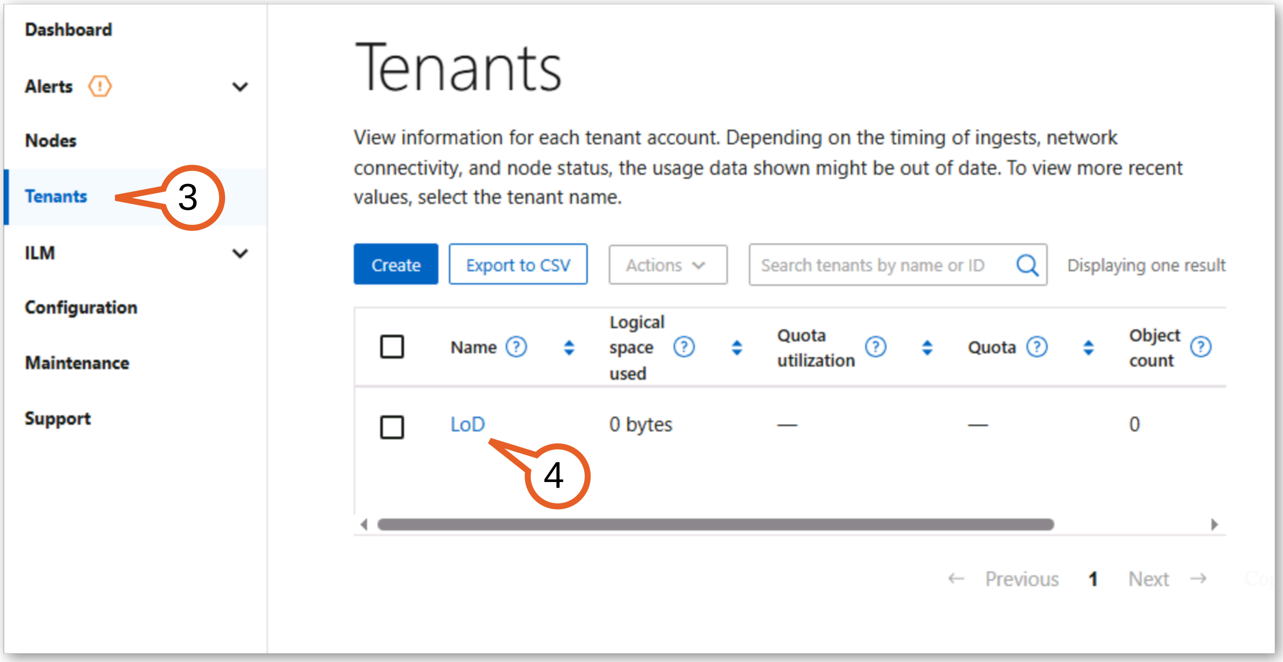

Figure 2-11:

5. Click **Sign in**.

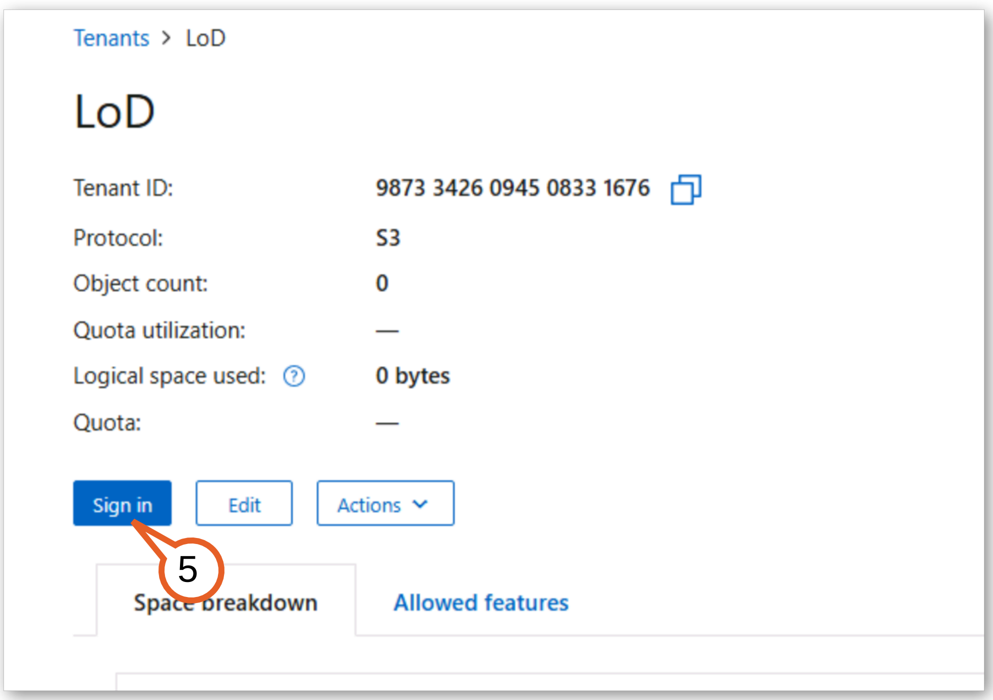

Figure 2-12:

6. Enter these credentials:
    - Username: `root`
    - Password: `Netapp1!`

    Leave the remaining fields unchanged and click **Sign in**.

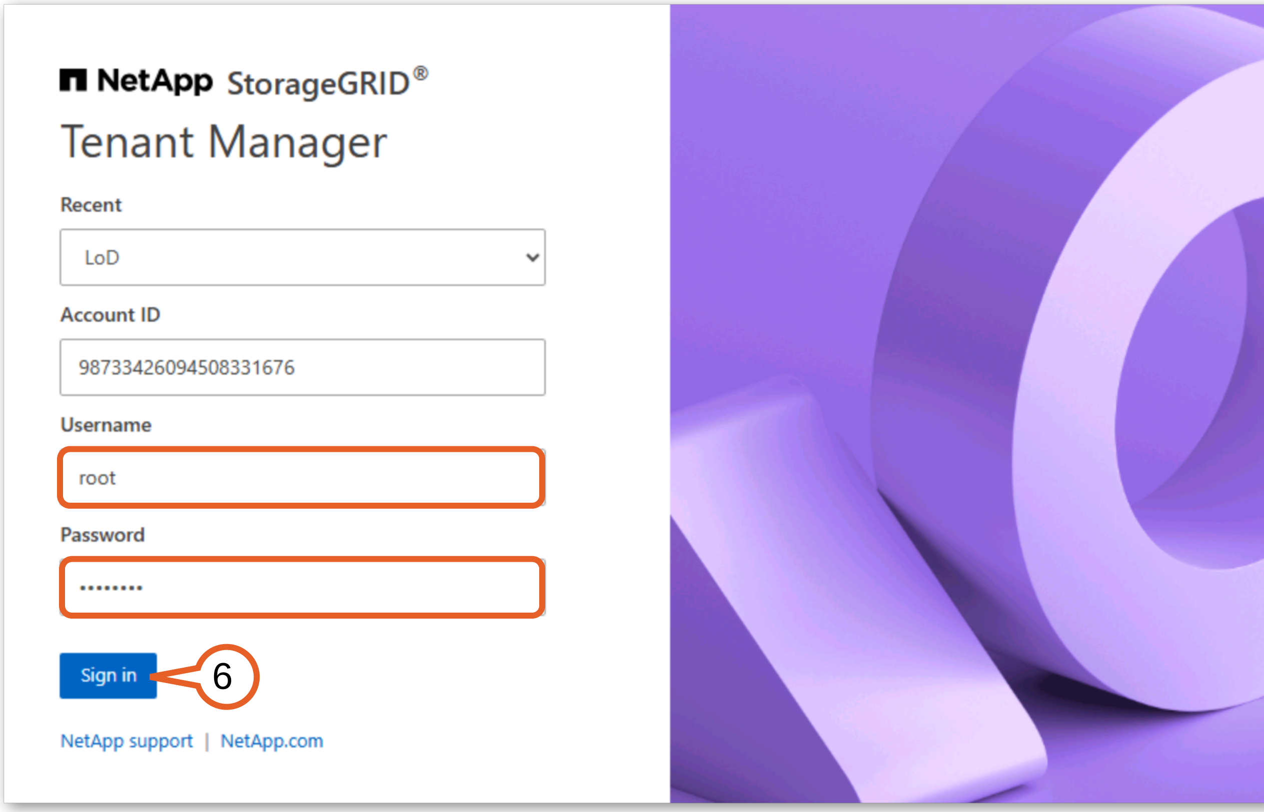

Figure 2-13:

7. Navigate to **Storage (S3)** > **Buckets**.

8. Click **Create bucket**.

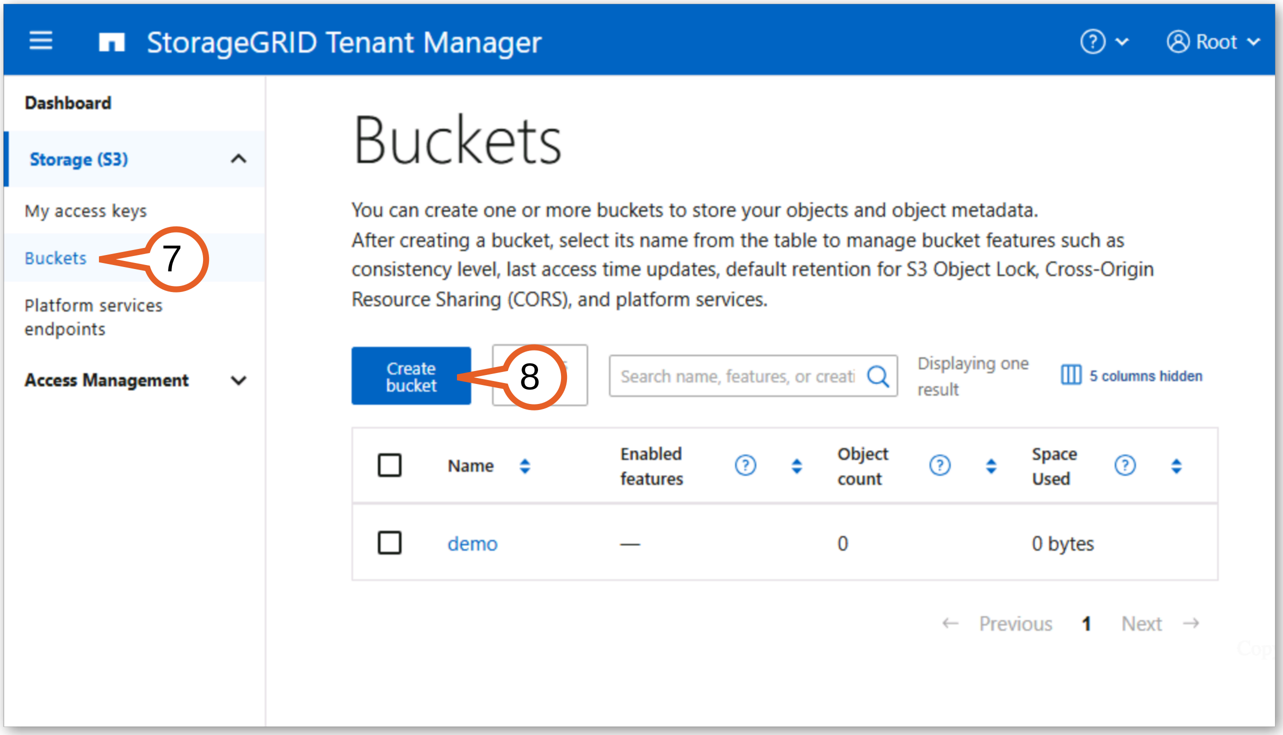

Figure 2-14:

9. In the Bucket name field, enter **`cluster2-tiering`**.

10. Click **Continue**.

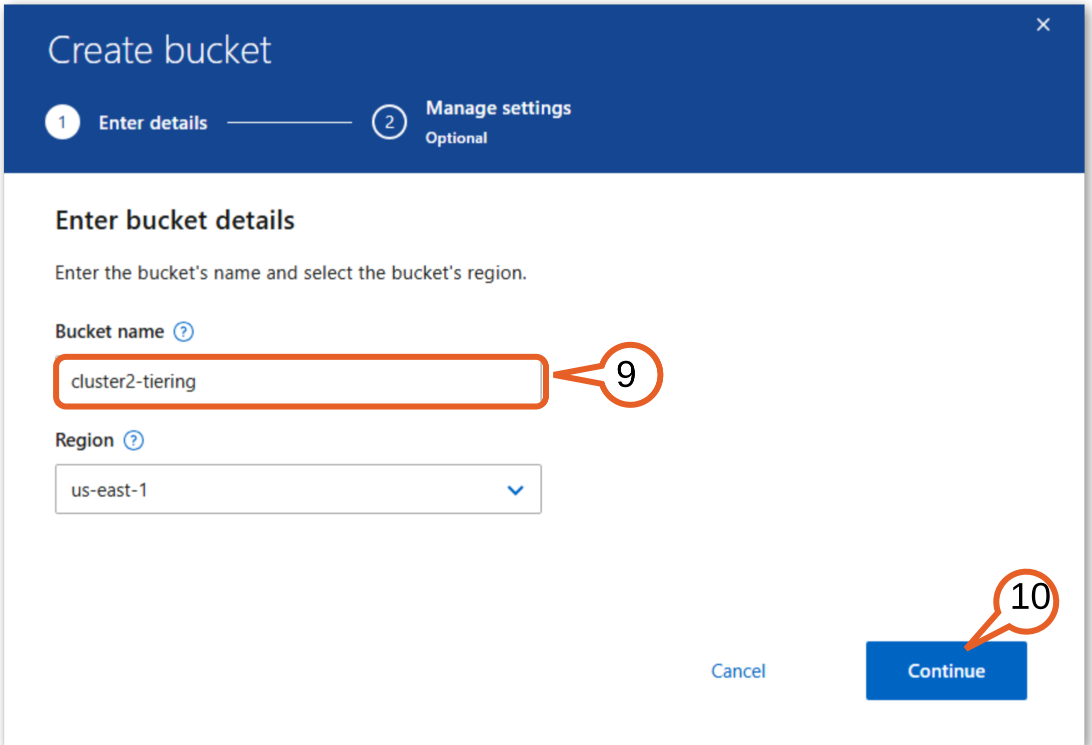

Figure 2-15:

11. Leave the remaining settings at their defaults, scroll down and click **Create bucket**.

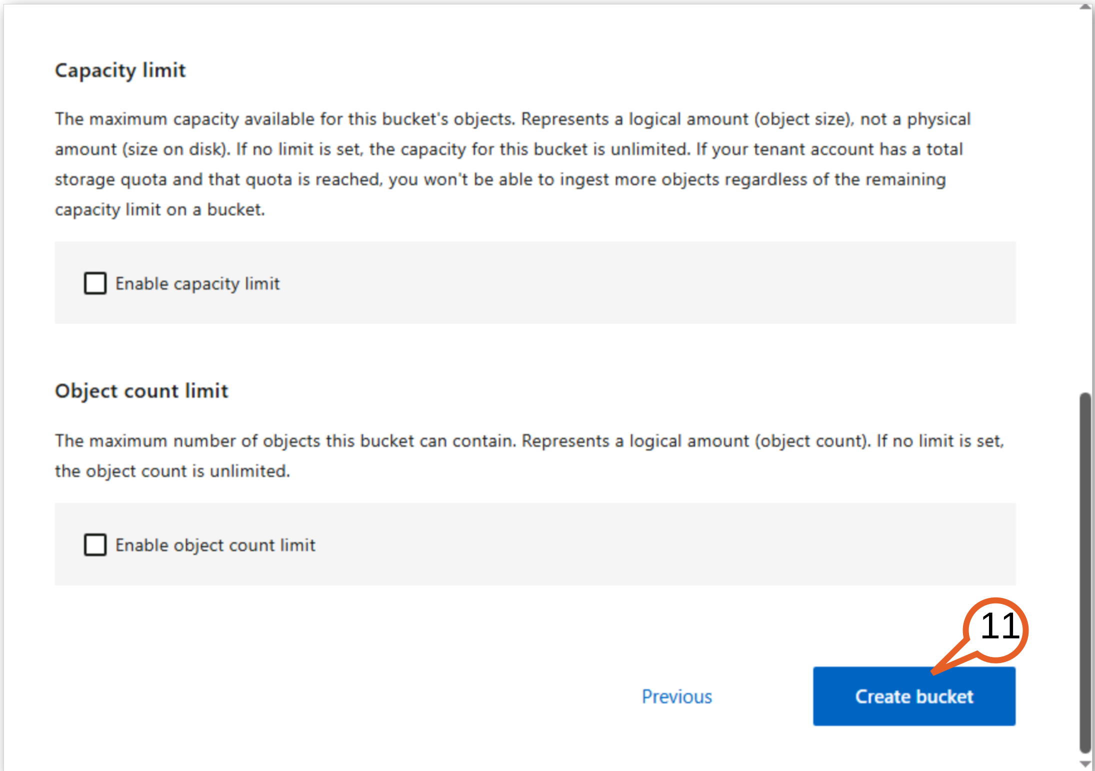

Figure 2-16:

12. Click **Finish**.

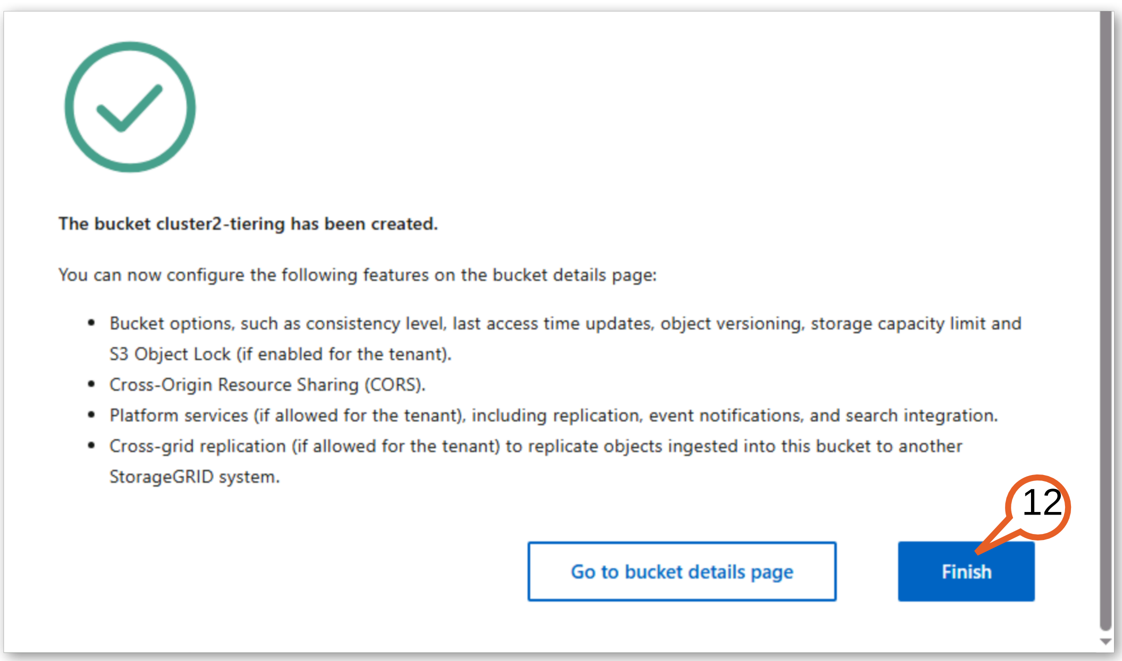

Figure 2-17:

13. Verify the new bucket, “cluster2-tiering” appears in the bucket list.

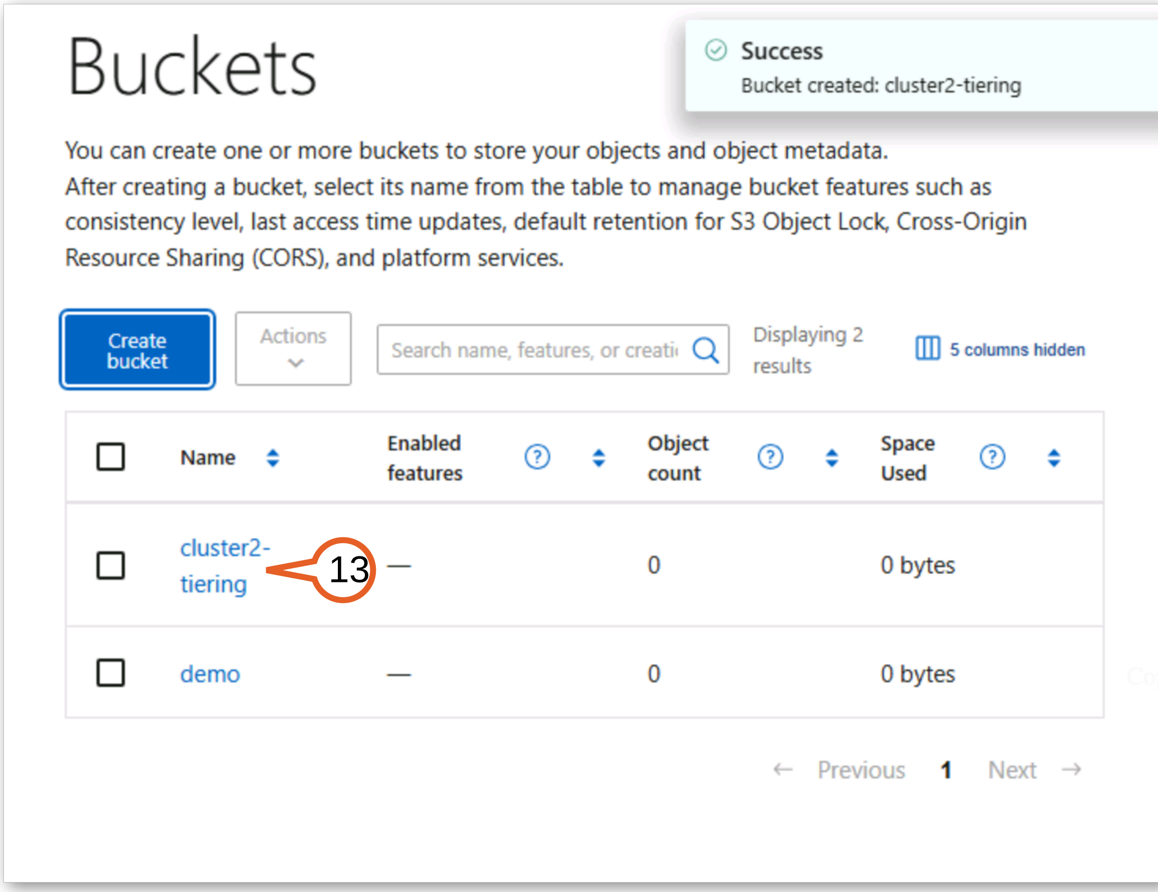

Figure 2-18:

Note: The S3 access key and secret key are also required to connect ONTAP to this bucket. You will use these in the next activity.

With the StorageGRID bucket created, you are now ready to attach it to your ONTAP storage tier.

2.1.4 Add a StorageGRID cloud tier

With the intercluster LIF and bucket in place, you can now register StorageGRID as a cloud tier in ONTAP. This step provides System Manager with the endpoint URL, S3 credentials, and server certificate needed to communicate with the object store. Once added, the cloud tier is available for attachment to any local tier on the cluster.

1. Click **Cluster2** in the Chrome browser bookmarks bar to return to System Manager for Cluster2.

   Note: If signed out, you will need to sign in again with username **`admin`** and password **`Netapp1!`**.

2. From the menu on the left, navigate to **Storage** > **Tiers**.

3. Click **Add cloud tier**.

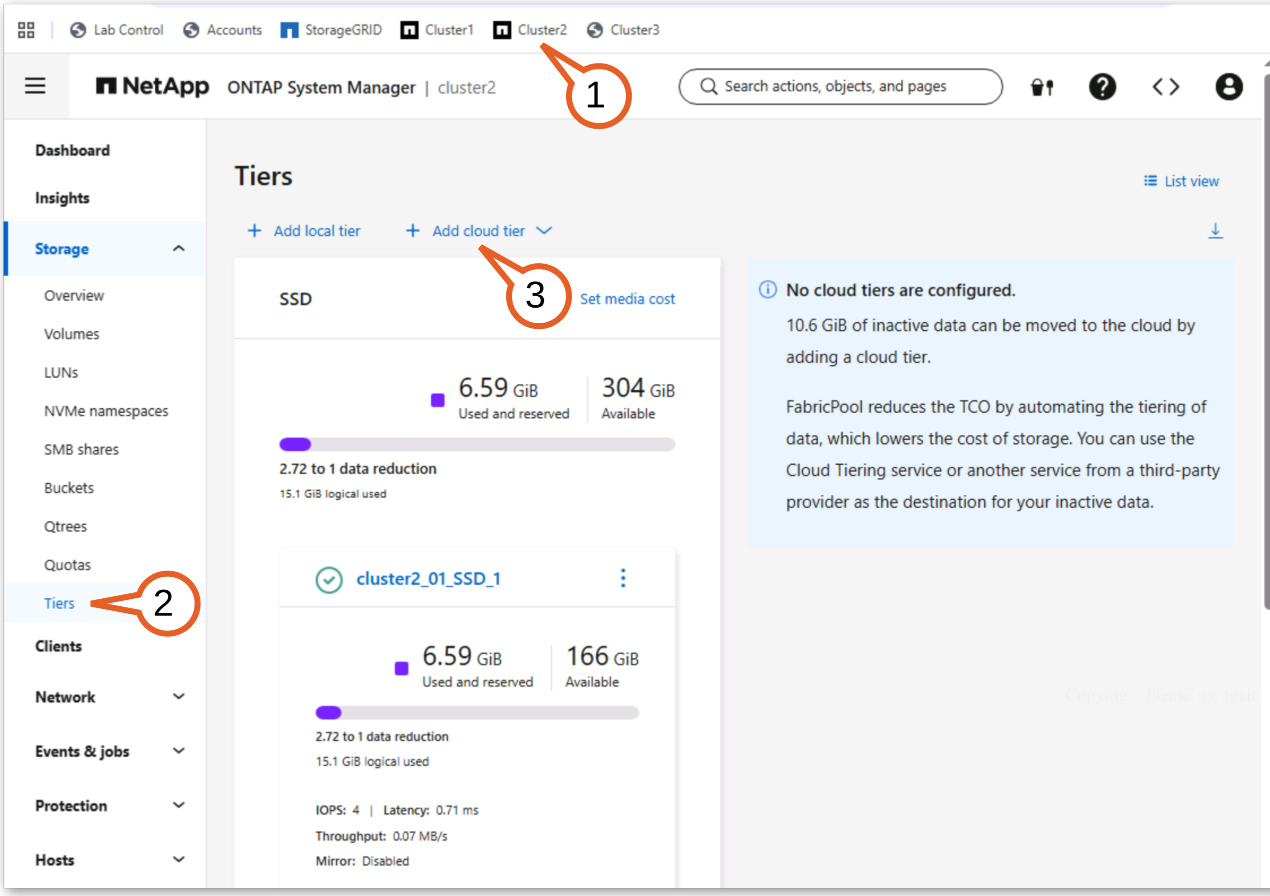

Figure 2-19:

4. From the **Object Store Provider** drop-down menu, select **StorageGRID**.

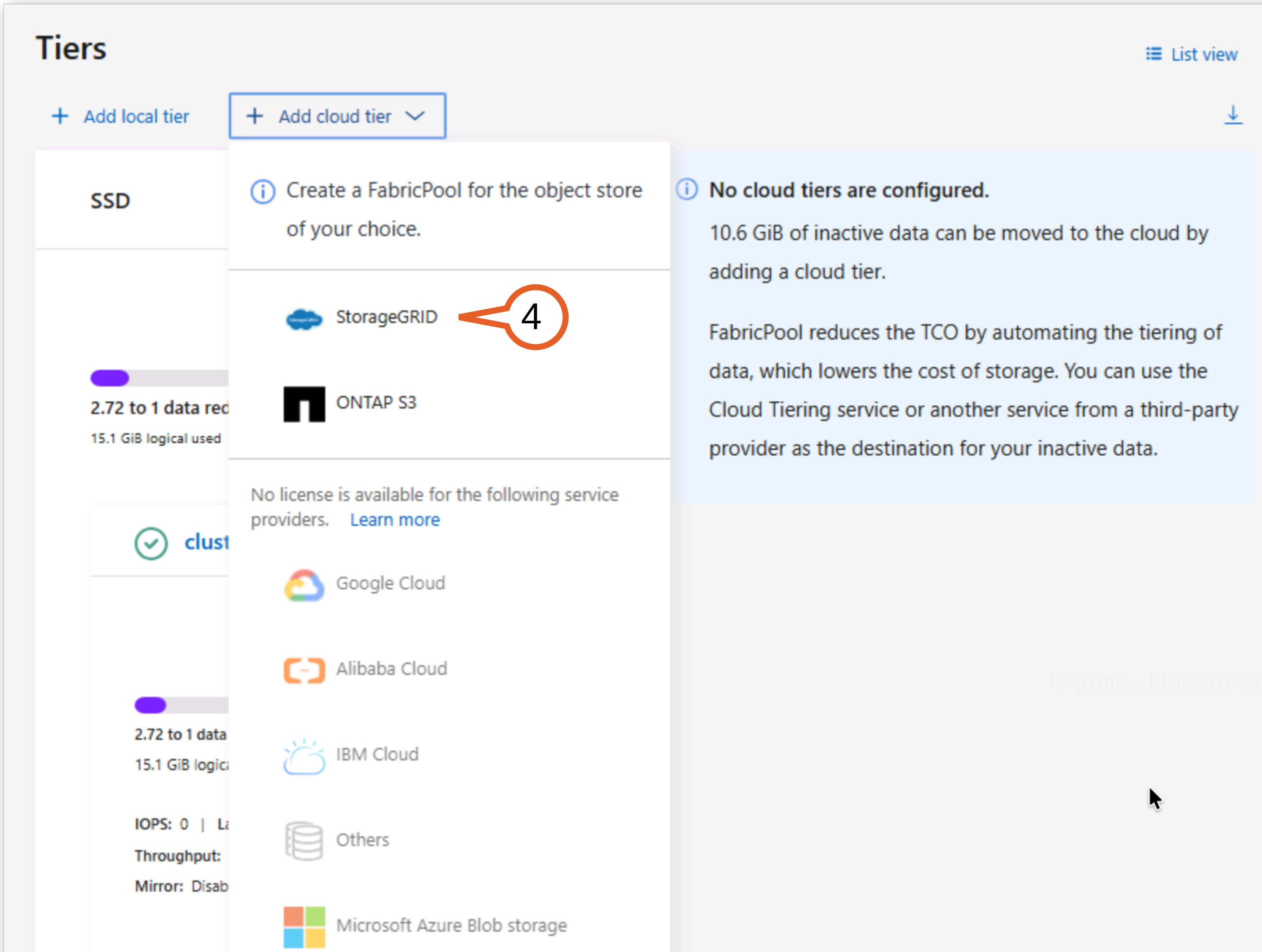

Figure 2-20:

5. Begin filling out the Add cloud tier form:
   - Server Name (FQDN): **`dc1-adm1.demo.netapp.com`**
   - Uncheck **SSL**.
   - Port: **`10080`**

Note: SSL encryption is recommended for using FabricPool, but HTTP is used in this lab to simplify the setup process.

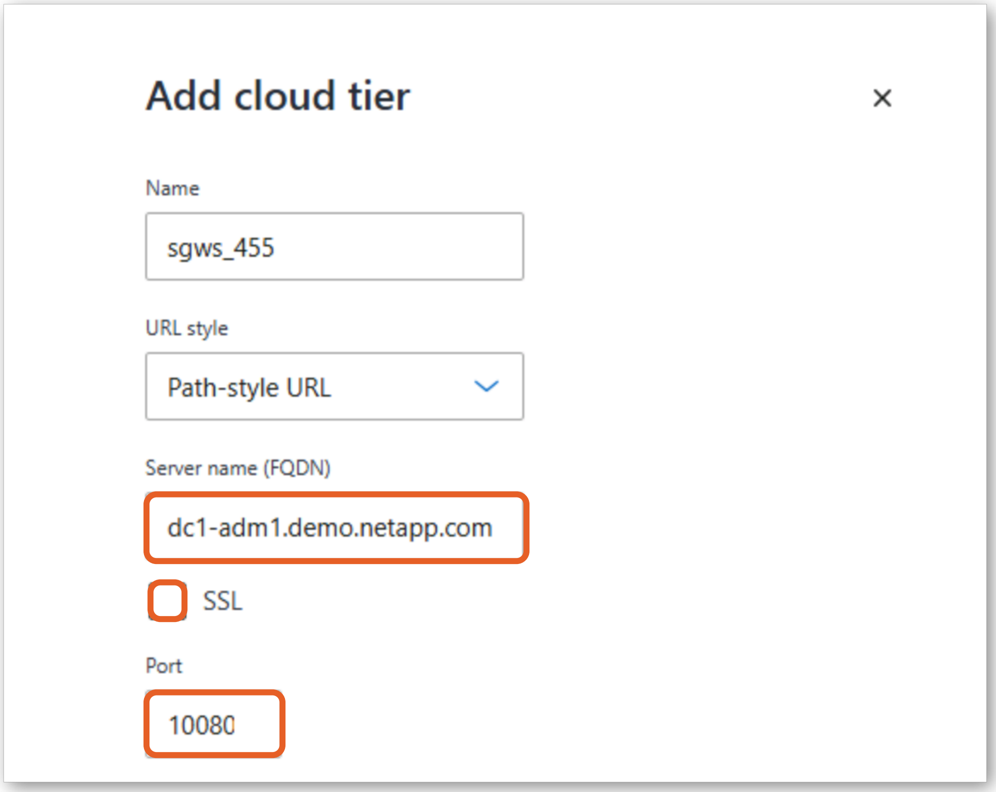

Figure 2-21:

6. Right click on **Accounts** in the Chrome bookmarks bar and click **Open in new tab**.

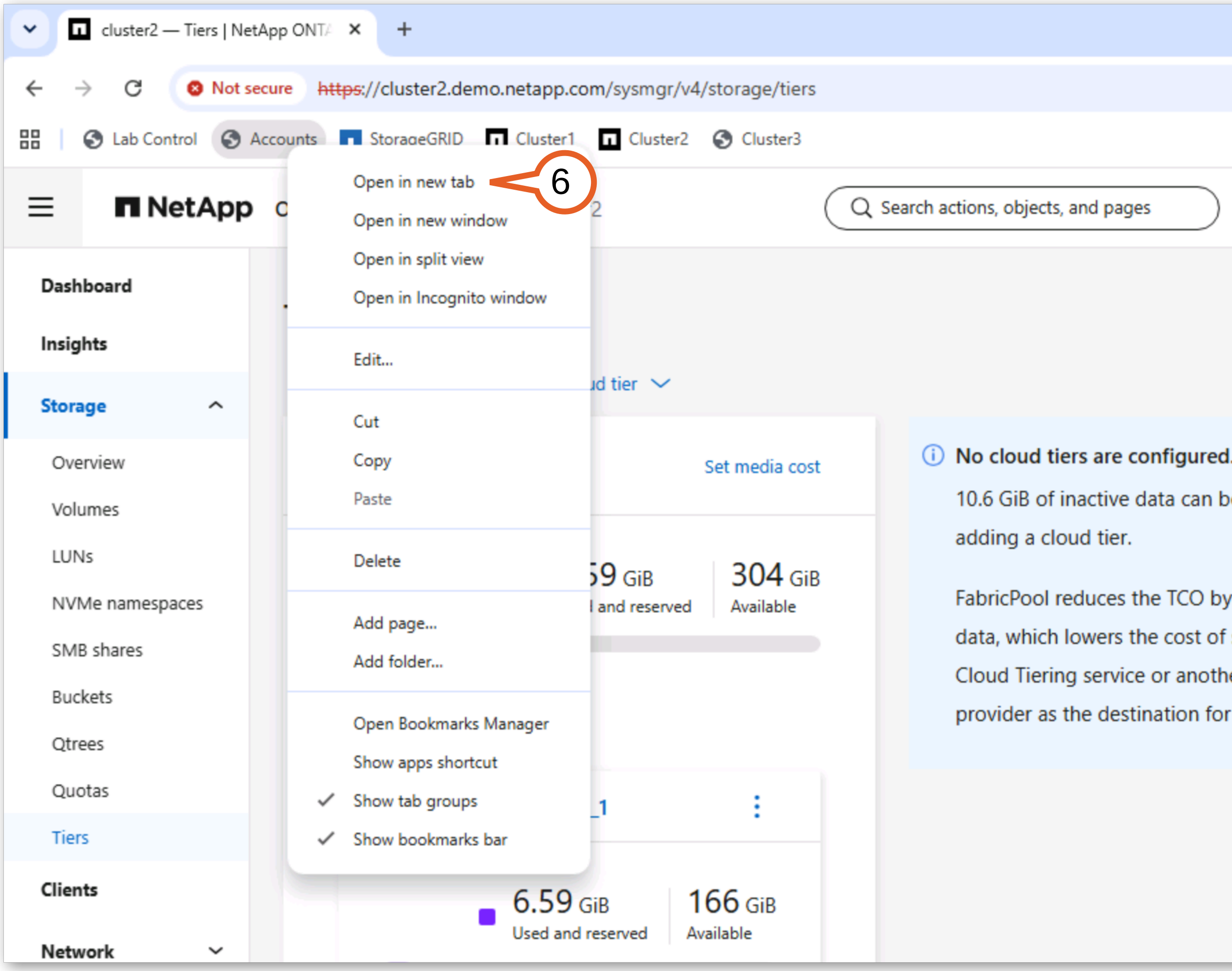

Figure 2-22:

7. On the Accounts page, locate the StorageGRID S3 access key and secret key listed under the StorageGRID credentials.

   Using the **copy buttons**, copy and then paste the Access key and Secret key credentials from the Accounts page into the “Add cloud tier” form.

8. Enter the bucket name **`cluster2-tiering`**.

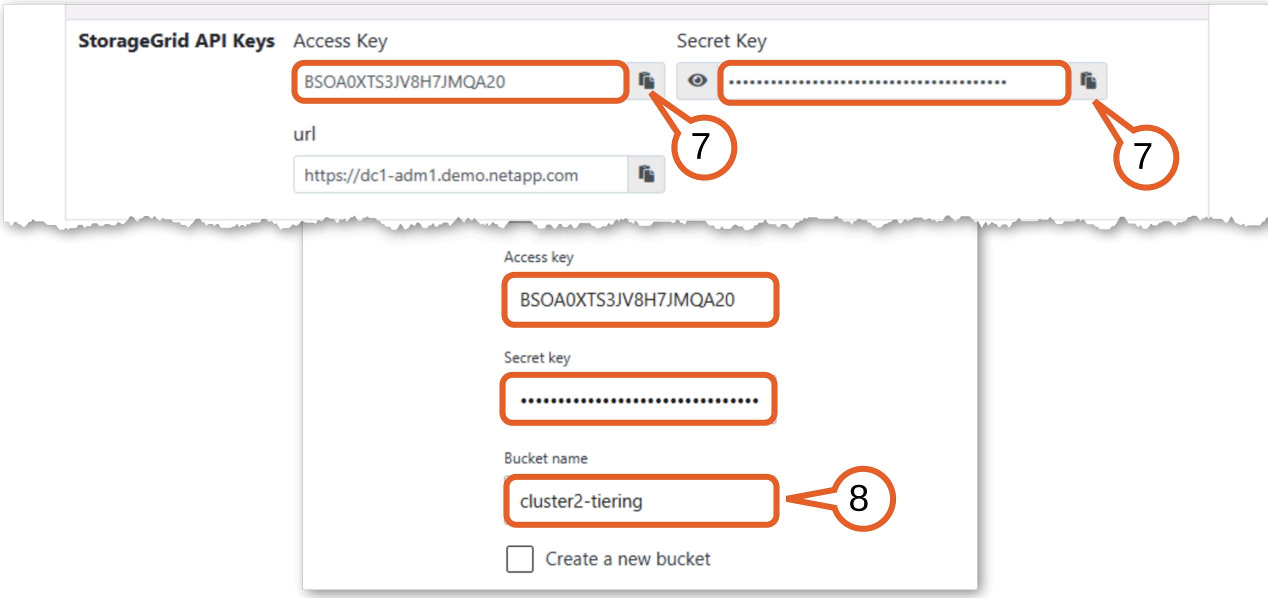

Figure 2-23:

9. Click **Save**.

Note: You could attach this new cloud tier to local tiers from within this view but you will do that as a later step. Also notice that the intercluster LIF is automatically selected in this view.

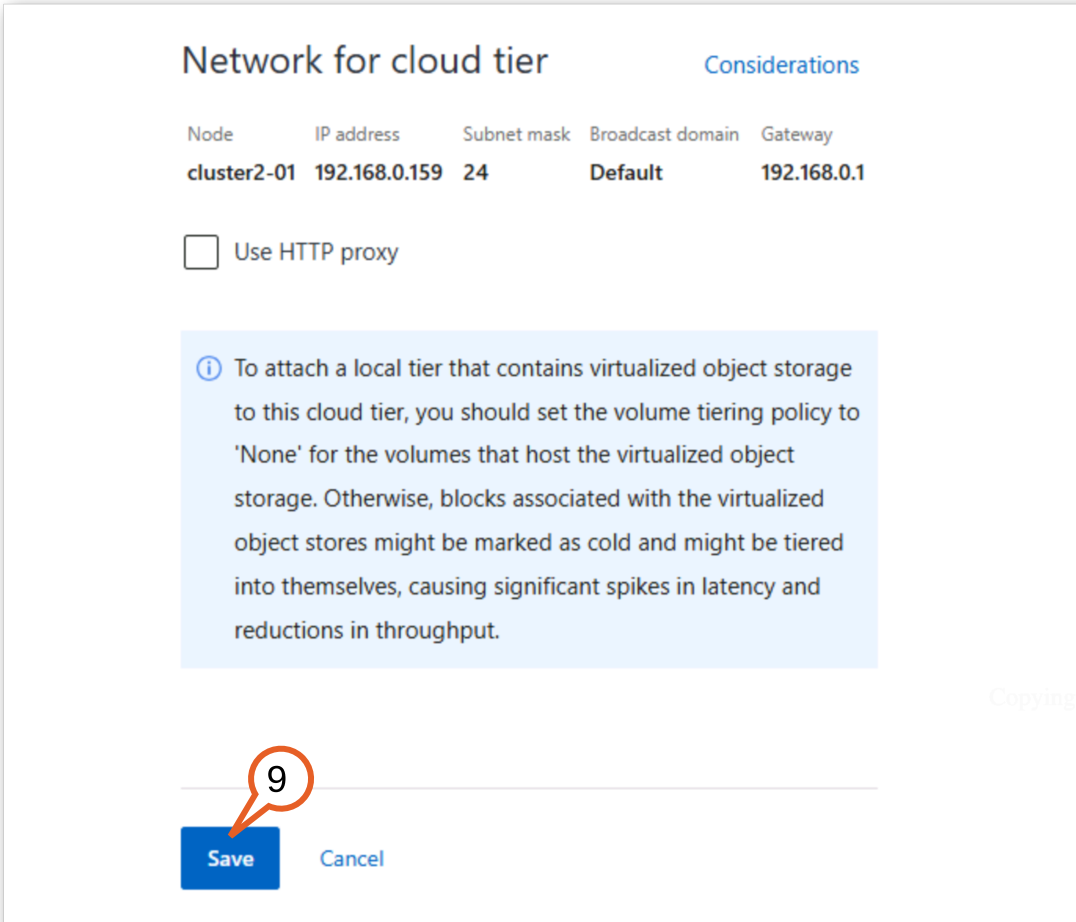

Figure 2-24:

10. Verify the new cloud tier appears in the Tiers view under "Cloud Tiers".

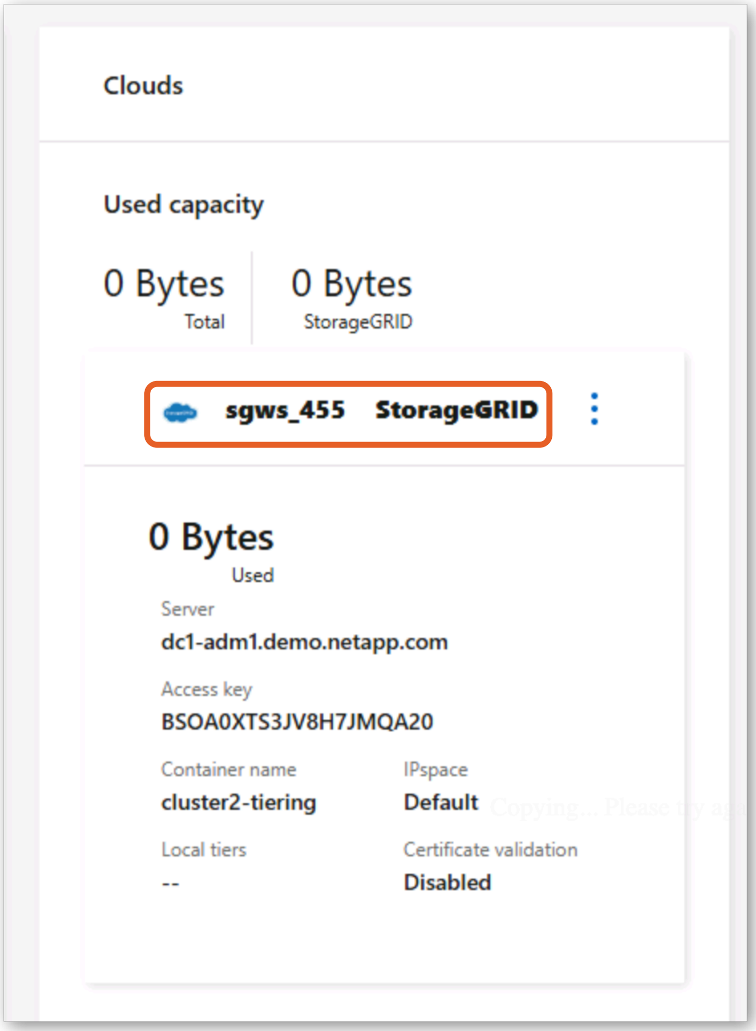

Figure 2-25:

Takeaway: Adding a cloud tier is a cluster-wide configuration: once registered, any local tier on the cluster can use it. The typical configuration uses HTTPS with certificate validation, ensuring data in transit is encrypted and the endpoint identity is verified.

2.1.5 Attach the cloud tier to a local tier

FabricPool uses a tiering policy applied at the volume level to decide which volumes on a FabricPool-enabled aggregate can participate in tiering. The tiering policy also specifies what types of data blocks within participating volumes are eligible for relocation to the cloud tier. Policies based on inactive data (Auto and Snapshot-Only) are coupled with a cooling period, measured in days, which controls how aggressively FabricPool moves data to the cloud tier.

ONTAP supports four volume tiering policies:

- **Snapshot-Only**—Cold Snapshot blocks in the volume that are not shared with the active file system are moved to the cloud tier. If read, cold data blocks on the cloud tier become hot and are moved back to the local tier.
- **Auto**—Moves all cold blocks in the volume to the cloud tier, not just blocks associated with Snapshot copies. If read by random reads, cold blocks become hot and return to the local tier. Sequential reads (such as index or antivirus scans) do not promote blocks back to the local tier.

- **All**—Immediately tiers all user data blocks regardless of access pattern. Blocks remain on the cloud tier even when read. Used for archival volumes and required for Cloud Write.
- **None**—Disables tiering for the volume. Any blocks previously moved to the cloud tier remain there until accessed, at which point they are restored to the local tier.

When a cloud tier is attached to an aggregate, FabricPool begins tiering data based on whatever tiering policy is in place for each volume on that aggregate. In this activity, you attach the StorageGRID cloud tier to the SSD local tier on Cluster 2 and examine the existing tiering policies.

1. You should still be in the Storage Tiers view of Cluster2 in your lab. Locate the local tier **cluster2_01_SSD_1**. Click the **menu icon** on the right side of the tier row and select **Attach Cloud Tier**.

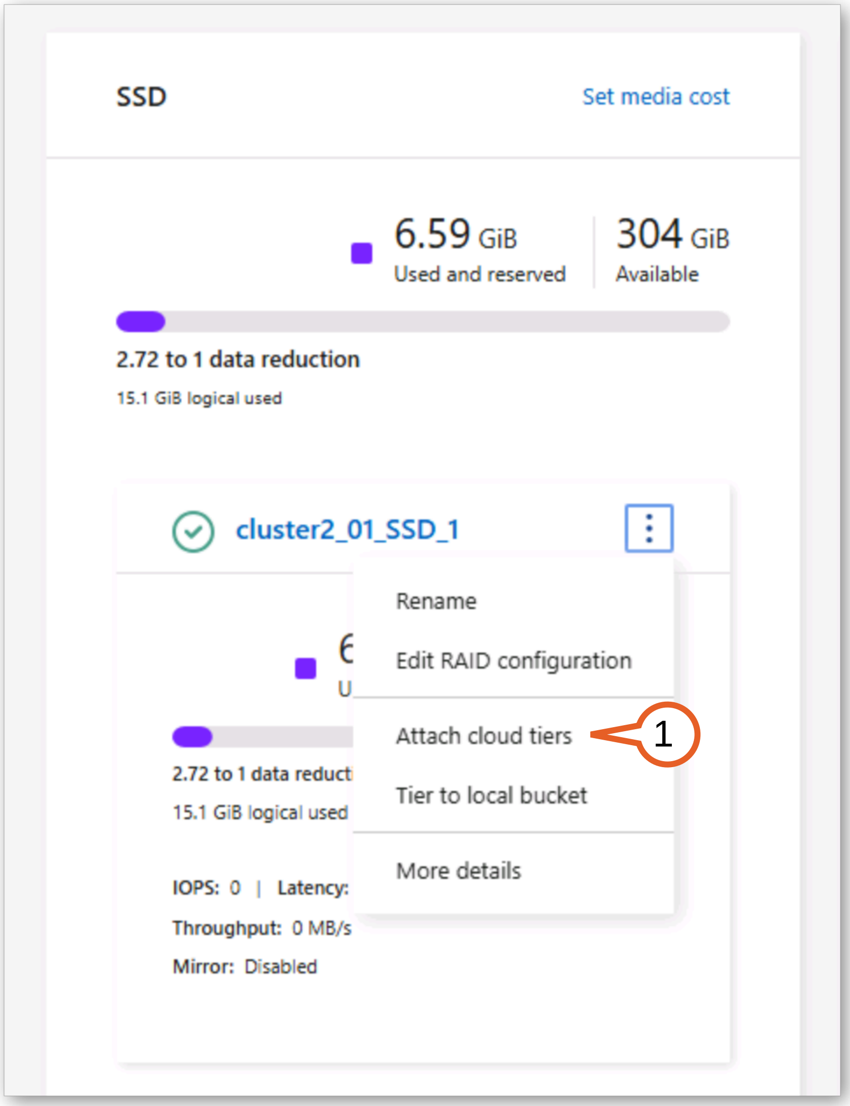

Figure 2-26:

2. In the Attach Cloud Tier dialog, you should see the cloud tier that you just created (the StorageGRID object store) selected.

3. Select **Update volume cloud tier properties**.

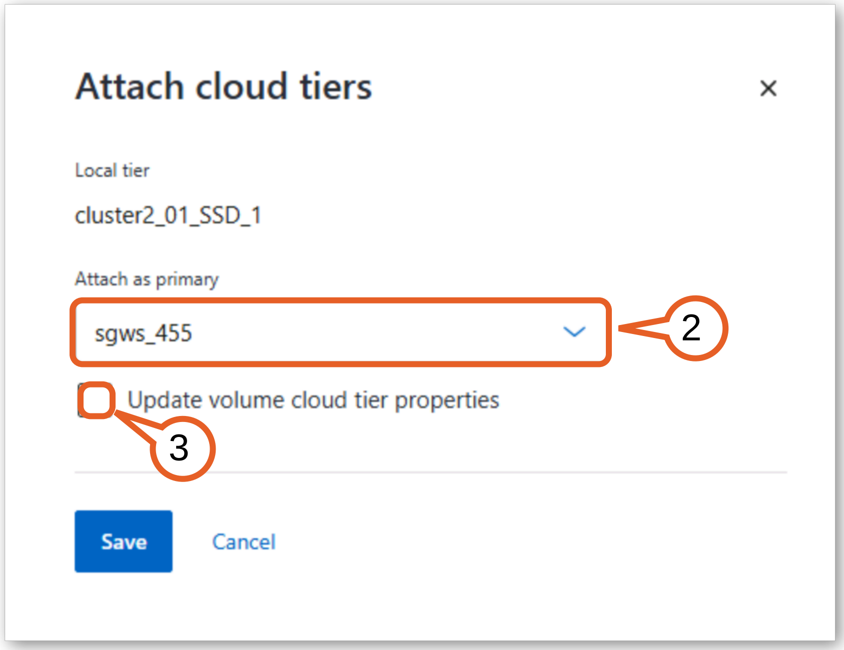

Figure 2-27:

4. From here you can change a volume’s assigned tiering policy using the drop-down menus in the table. All of these volumes have already been configured with a tiering policy of Auto.

   In a later activity, you will configure the tiering policy for a volume. For now, leave the volumes with their current selections.

   Click **Save**.

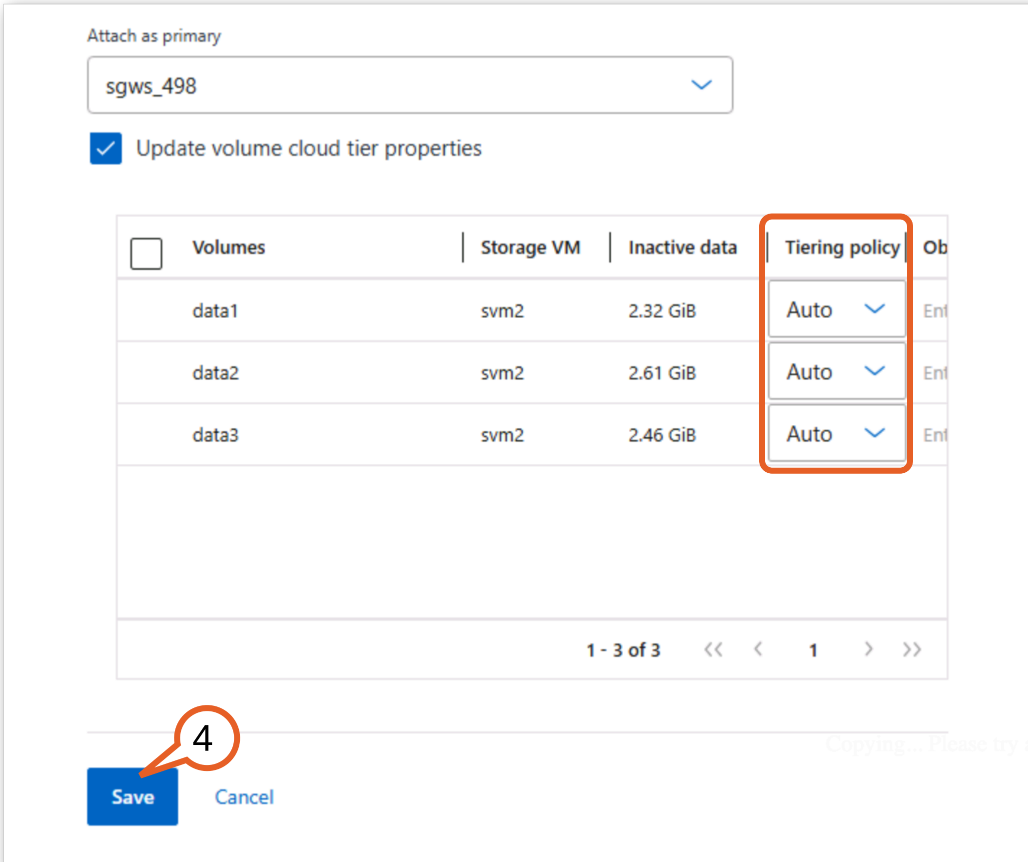

Figure 2-28:

5. In the confirmation pop-up that appears, click **Ok**.

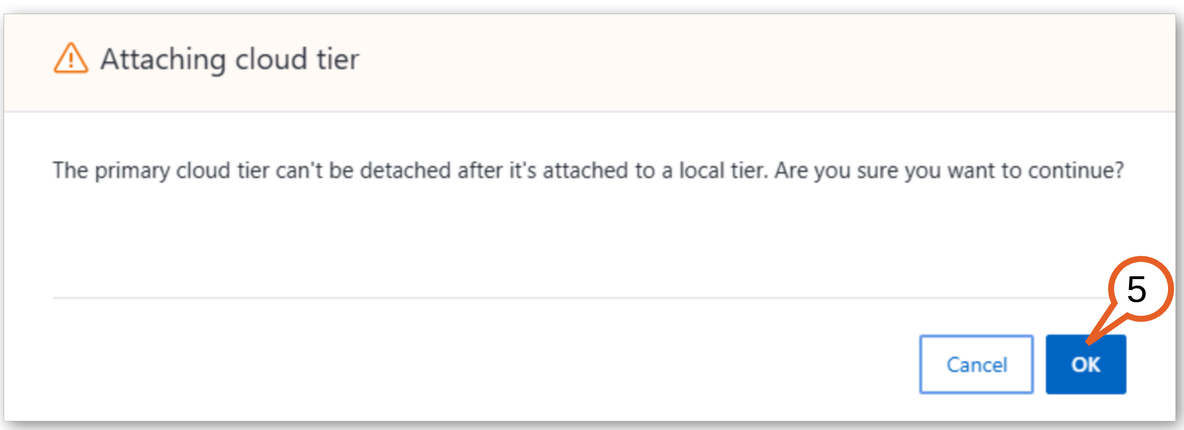

Figure 2-29:

6. Verify that the local tier now displays a cloud tier indicator showing the attached StorageGRID object store.

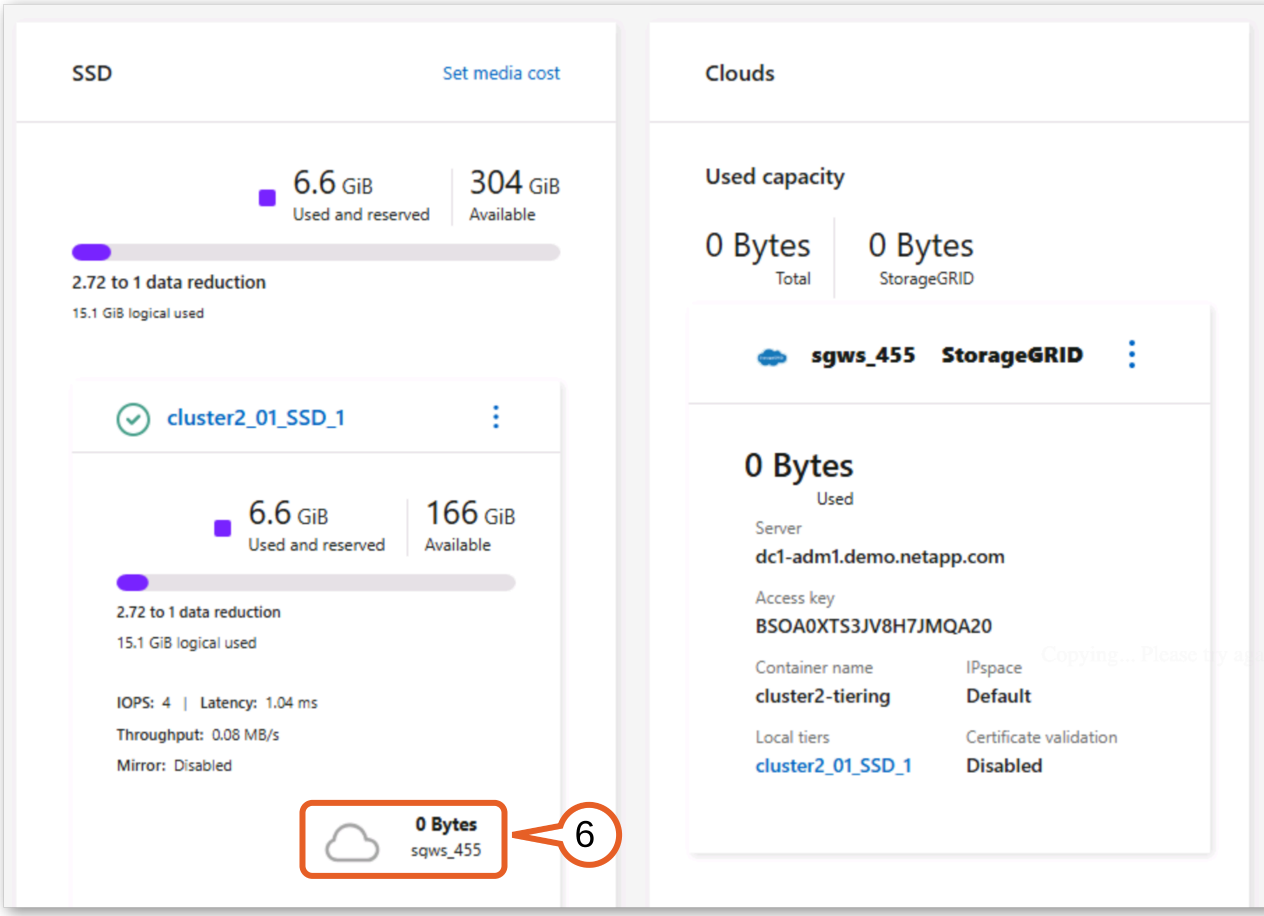

Figure 2-30:

FabricPool will not automatically relocate a volume's data blocks to a cloud tier unless the following criteria are met:

- The volume must reside on a FabricPool aggregate (an aggregate with an attached cloud tier).
- The volume's assigned tiering policy must permit some form of tiering.
- Eligible blocks in the volume must have gone unaccessed long enough to become cold.
- The aggregate must have crossed the tiering fullness threshold (default 50%; adjusted to 2% in this lab to simplify demonstration).

With all of these criteria met, you have completed the setup of FabricPool on ONTAP Cluster 2. This cluster is now ready to begin tiering after the next tiering scan. Rather than wait for that, for the rest of the lab you will switch to ONTAP Cluster 1 which already has data tiered to the cloud tier.

> **Takeaway:** Setting up on-prem FabricPool requires minimal effort, no additional licensing, and supports a variety of ONTAP configurations.

2.2 Manage Tiering

Despite working almost invisibly compared to manual tiering solutions, FabricPool allows you to have control over how data is tiered with its automated process. You can control things like how aggressively you want to tier data from a given volume, and you can manually promote specific data to the performance tier without having to wait for the data to be accessed first. These capabilities ensure that FabricPool will make decisions that do not conflict with your organization's objectives while still providing the benefits of automated tiering.

Important: You now shift to a production cluster that already has FabricPool configured and data tiered to an object store. This pre-configured environment lets you focus on analyzing and managing tiered data rather than repeating setup steps.

In the following activities, you inspect tiering results, change a volume's tiering policy, and write data directly to the cloud tier.

Activities
Analyze tiering results
Changing a Volume’s Tiering Policy
Enable Cloud Write

2.2.1 Analyze tiering results

Cluster 1 has been tiering cold data to its ONTAP S3 cloud tier for some time. ONTAP 9.18.1 introduces enhanced space reporting that breaks down cloud tier usage into “referenced” and “unreferenced” capacity, giving you visibility into how much tiered data is still actively pointed to by volumes versus data that remains in the object store after volume deletions or overwrites. This distinction helps you identify reclaim opportunities and understand the true cost of your cloud tier usage.

Note: FabricPool's tiering mechanism works transparently across a wide range of object store targets, so you can pick the destination that best fits your cost, performance, sovereignty, and operational needs. On-premises options like NetApp StorageGRID and ONTAP S3 keep tiered data in your own data center, while public cloud targets such as Amazon S3, Microsoft Azure Blob Storage, and Google Cloud Storage offer elastic capacity. Cluster 1 tiers to an ONTAP S3 bucket, but the workflow and results are identical regardless of which supported object store backs the cloud tier.

1. In your browser, navigate to the **Cluster 1** bookmark to open System Manager.
2. Sign in with the following credentials:
    - Username: **`admin`**
    - Password: **`Netapp1!`**

3. Select **Keep me signed in for this session.** and click **Sign in**.

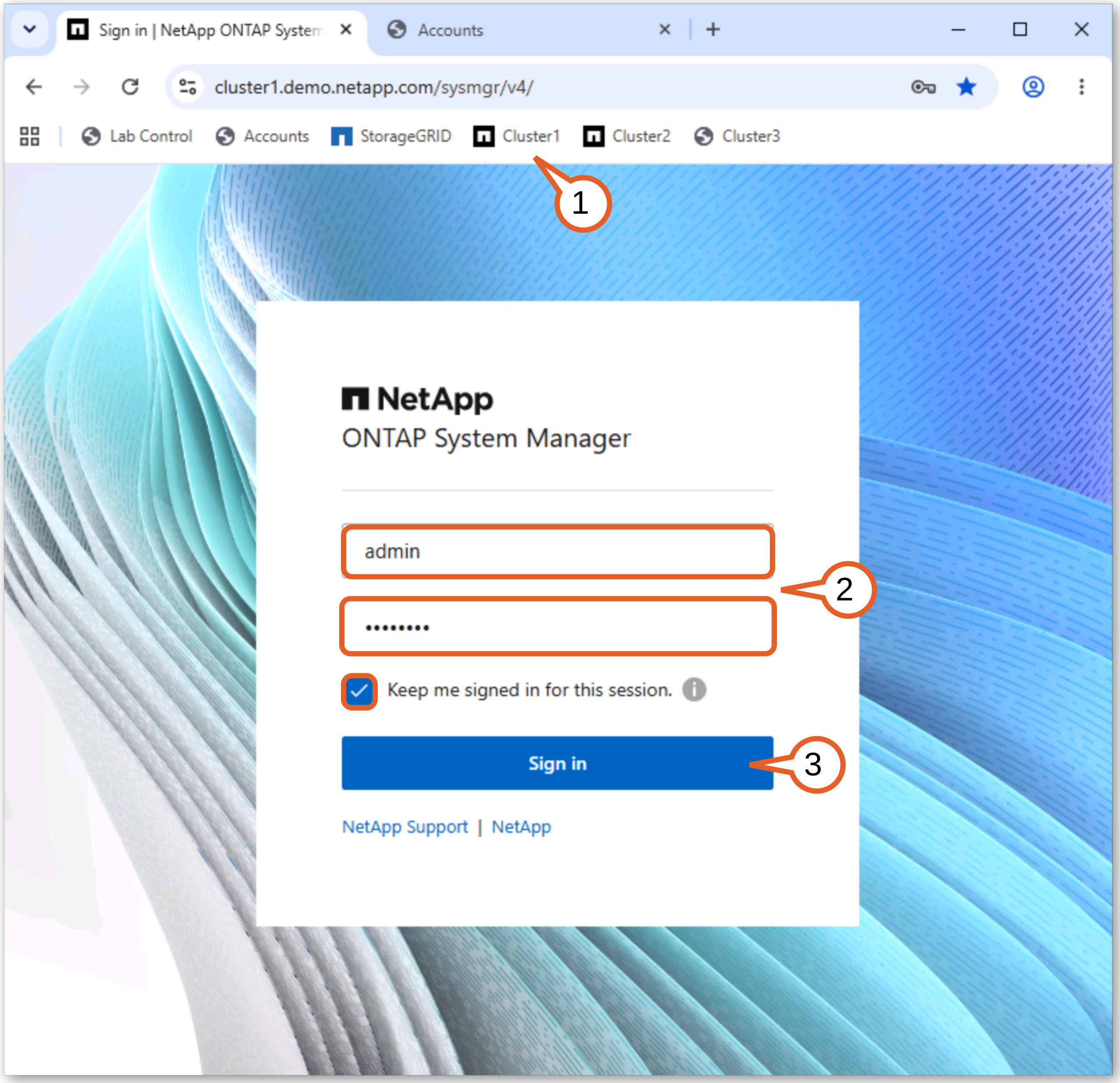

Figure 2-31:

4. Navigate to **Storage** > **Tiers**.

5. On the right side is a Clouds section. Here you will see a panel for an ONTAP S3 object store that has two local tiers attached and already contains around 7GB of tiered data.

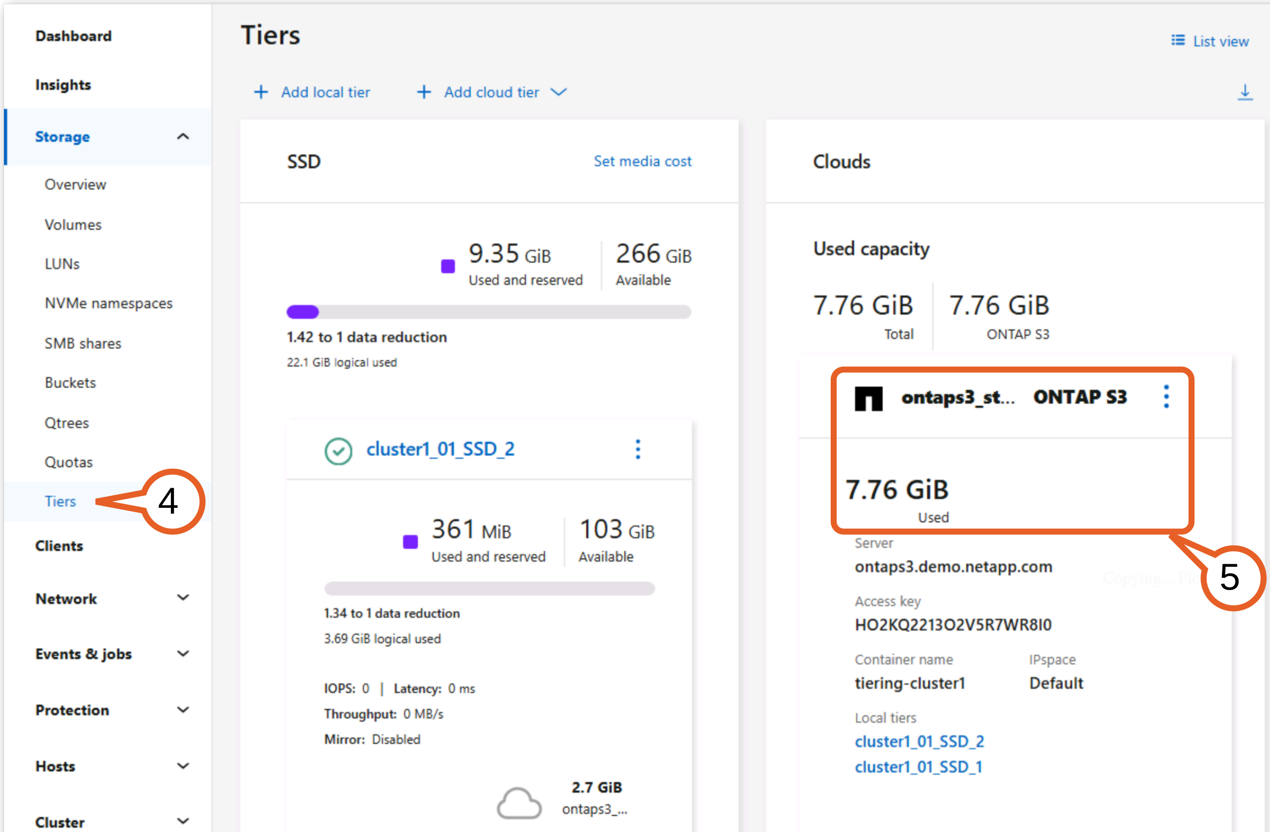

Figure 2-32:

6. In the SSD section on the left side is a list of the aggregates within this ONTAP cluster.

   Two of the three aggregates already have cloud tiers attached. The number next to the cloud icon indicates how much of the data belonging to the respective aggregate has already moved to that cloud tier. For all of the aggregates that have attached cloud tiers, the used capacity shown for the internal tier does not include data that has been moved to the cloud tier.

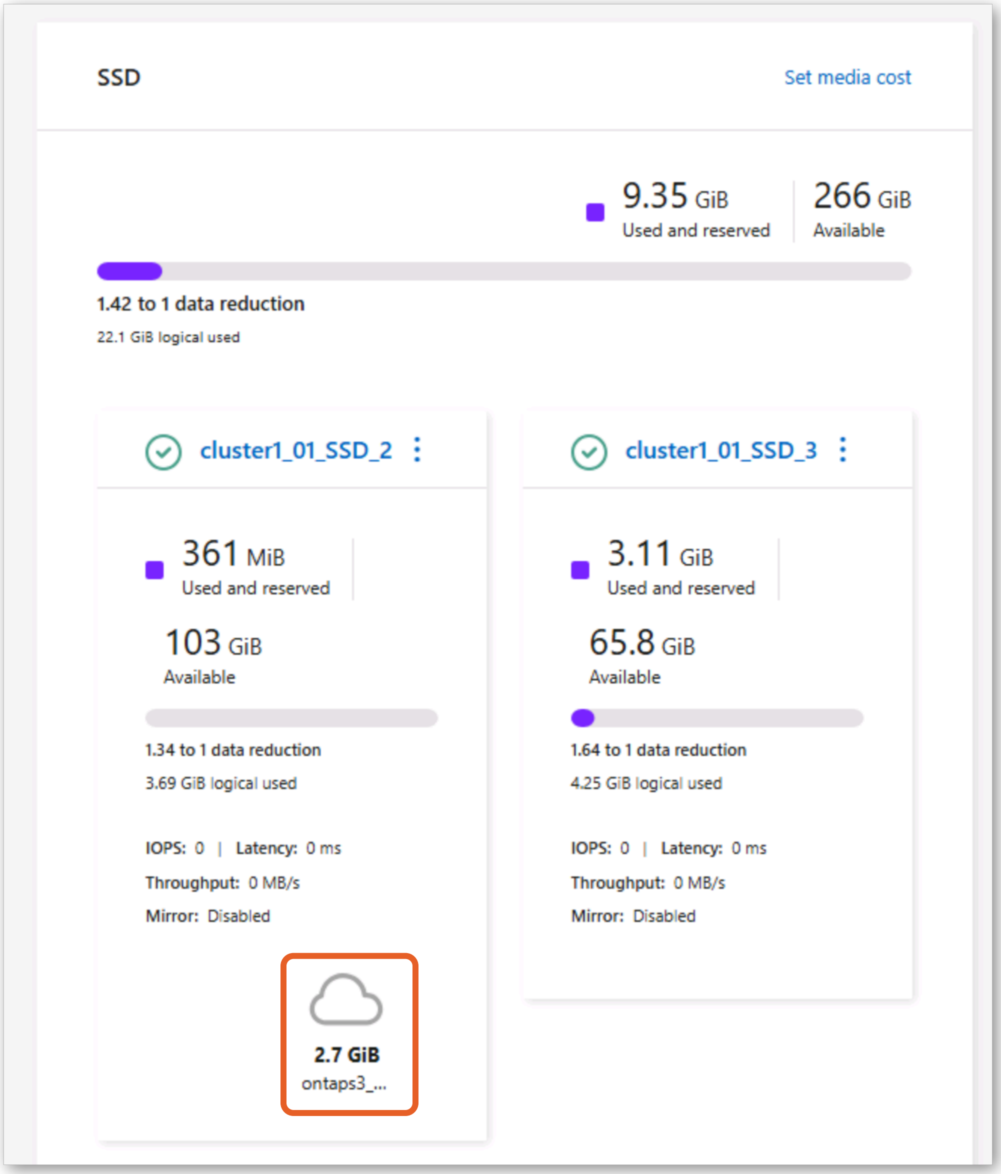

Figure 2-33:

Scroll down and locate the “cluster1_01_SSD_1” aggregate within the SSD pane. Observe that “cluster1_01_SSD_1” currently has about 5 GiB tiered out to the cloud, indicated by the counter in the bottom right of the box.

7. Click **cluster1_01_SSD_1**.

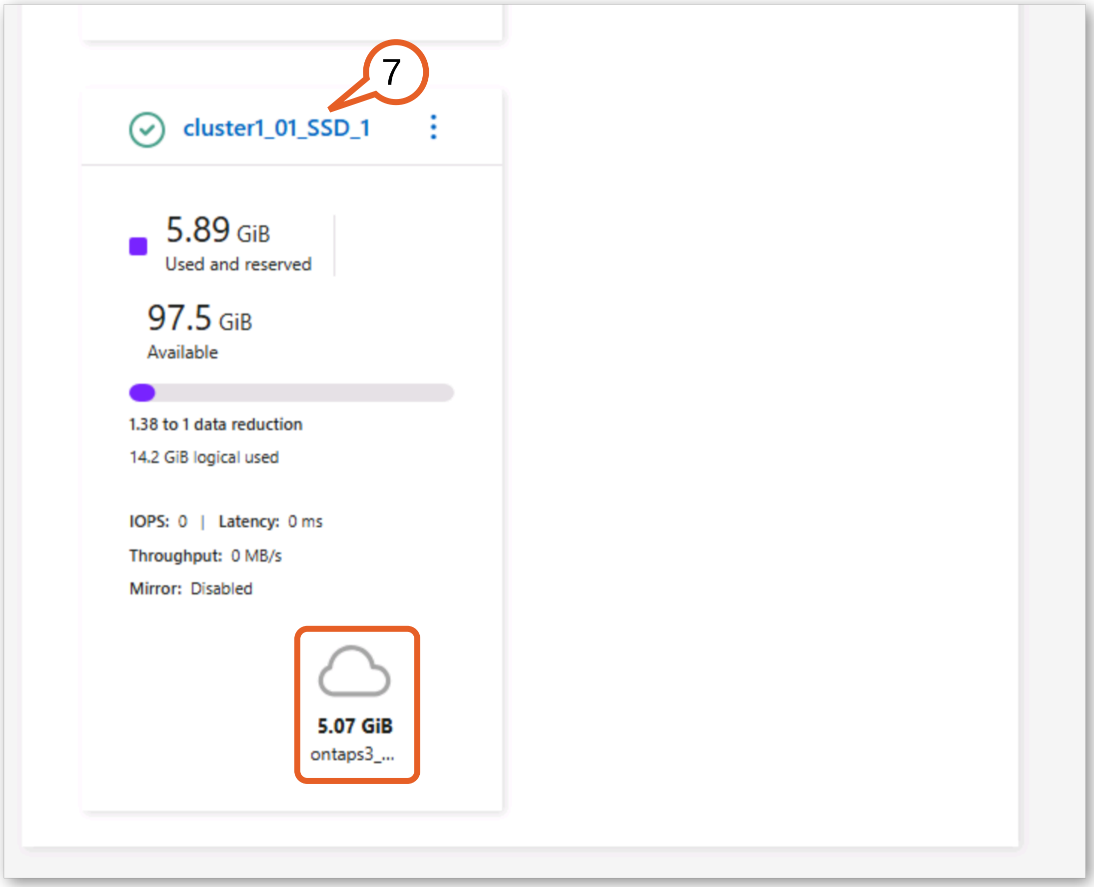

Figure 2-34:

8. On the Overview page for cluster1_01_SSD_1 you no longer see inactive data reported like you saw on Cluster 2 earlier in this lab. This is because that data has already been moved to the cloud tier. Instead you see the total amount of data tiered to the cloud like in the previous view.

Click the **Volumes** tab.

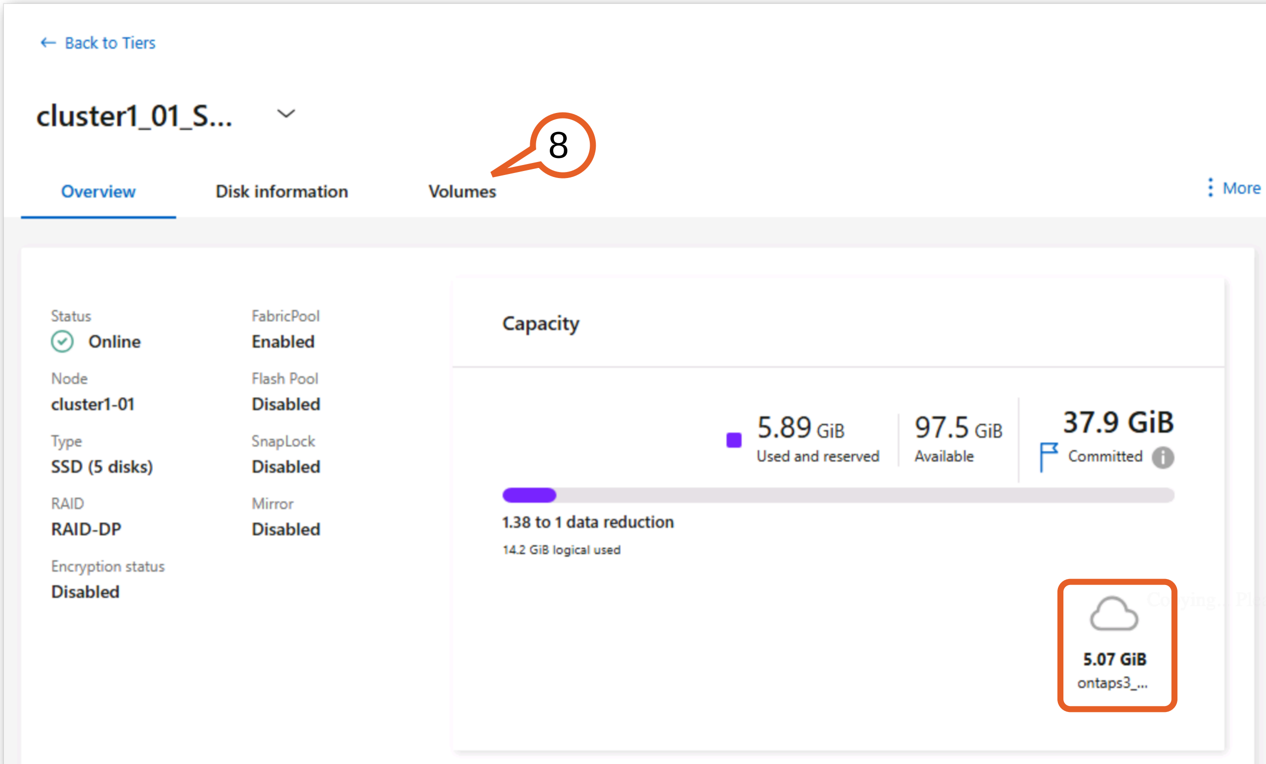

Figure 2-35:

9. You can use this table of volumes within “cluster1_01_SSD_1” to get an idea of how data is currently tiered as well as the current tiering policy and cooling period.

Observe that the “proj1” volume has 2.76 GiB in the local tier and 2.69 GiB tiered to the cloud using the Auto tiering policy to keep only warm data in the local tier.

10. This information is also available on the overview page for a given volume within System Manager. Click **proj3**.

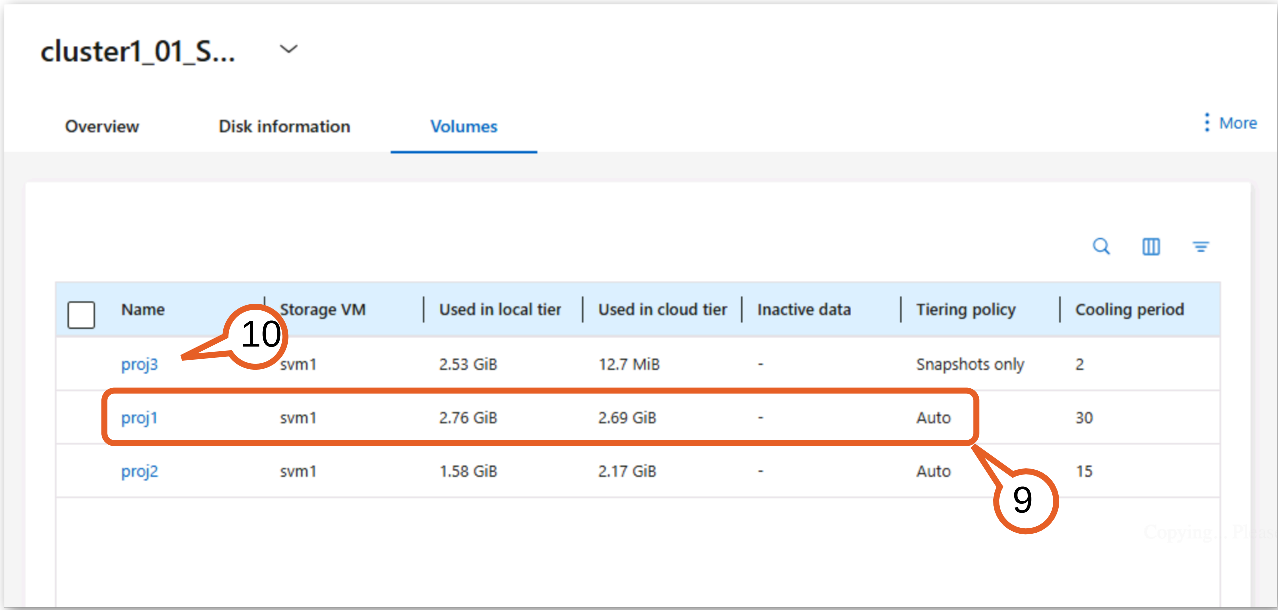

Figure 2-36:

11. Observe that the “proj3” volume is 10.5 GiB in size, and currently about 2.55 GiB are used.

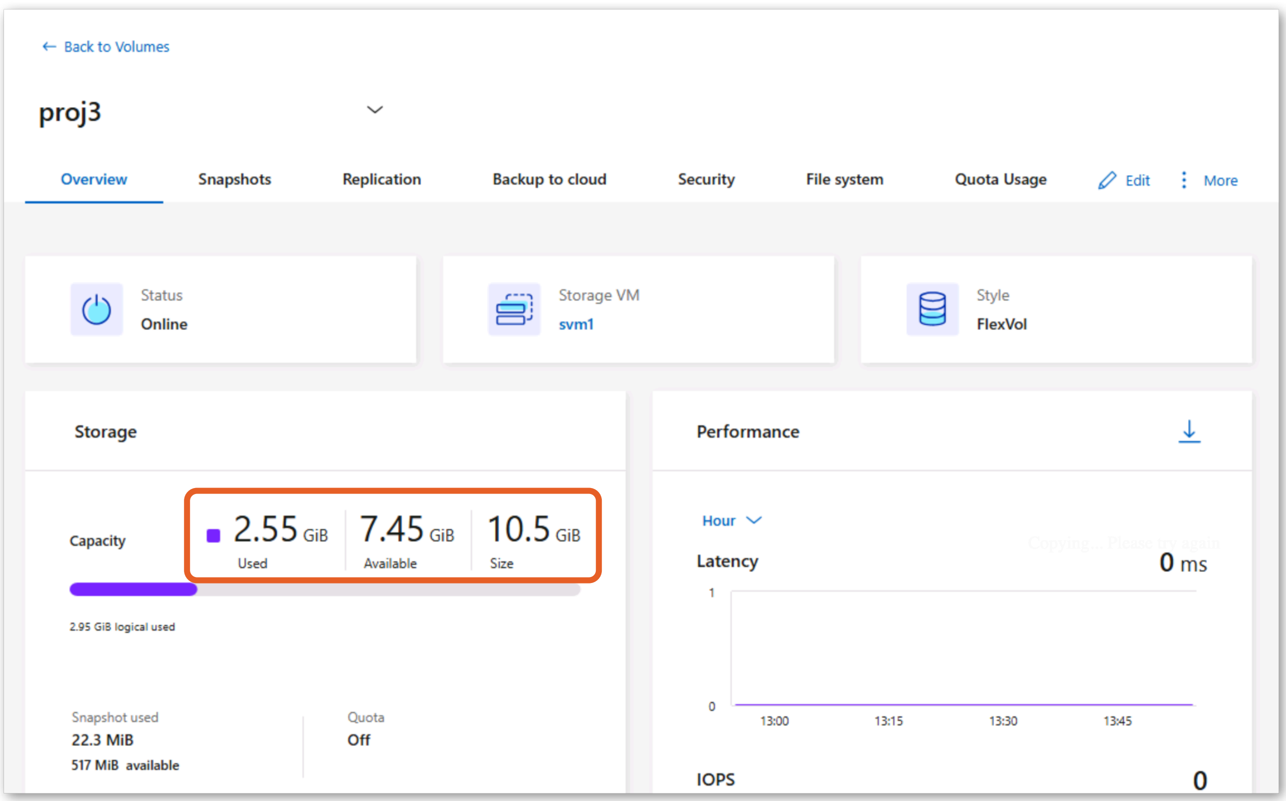

Figure 2-37:

12. Scroll down to the volume's Tiering section. Notice that the volume's tiering policy is set to "Snapshot_only" with a cooling period of 2 days, as indicated in the field on the left. This means that only cold blocks that are captured in snapshots (and that are not referenced by the active filesystem) are eligible for tiering.

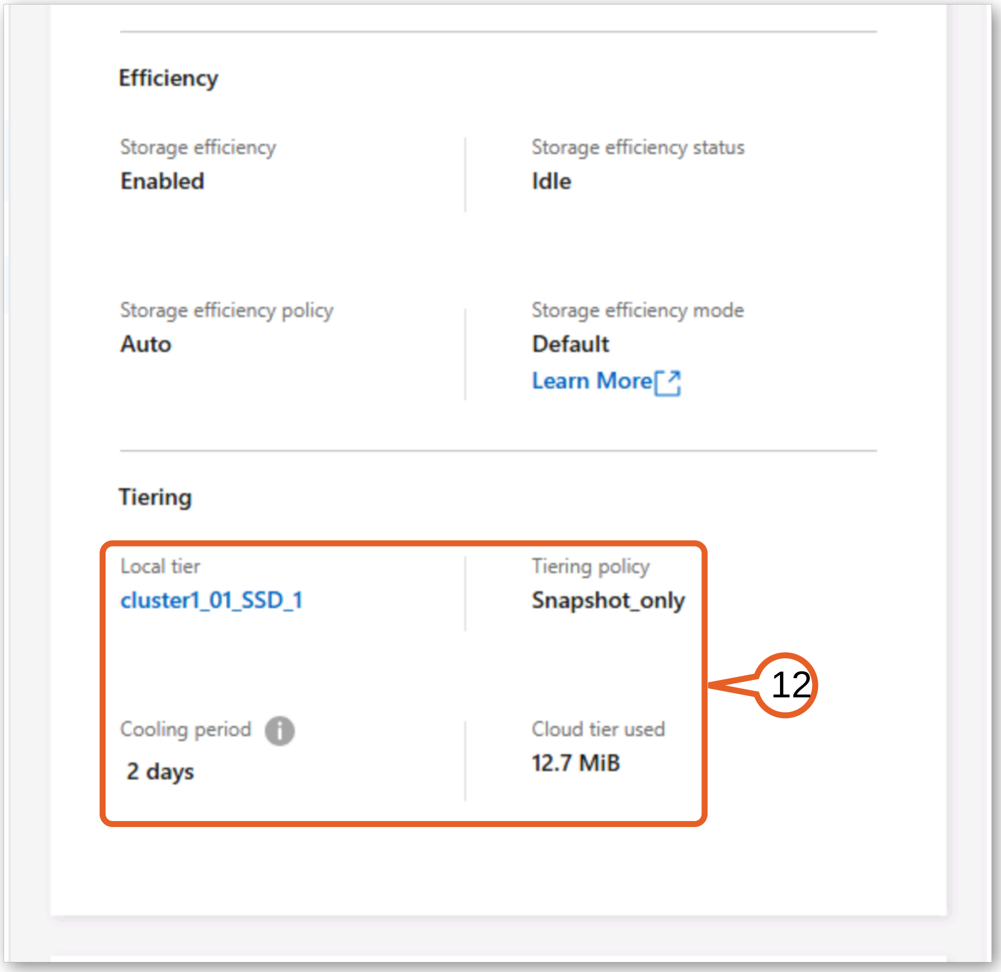

Figure 2-38:

Tip: You can also use the ONTAP CLI to quickly gather information about how much data is tiered and object-store utilization.

The `show-footprint` command presents a breakdown of a volumes footprint across storage types:

```
cluster1::> volume show-footprint -vserver svm1


      Vserver : svm1
      Volume  : proj1

      Feature                                          Used       Used%
      --------------------------------             ----------    -----
      Volume Data Footprint                            5.45GB        5%
             Footprint in Performance Tier             2.98GB       53%
             Footprint in ontaps3_store                2.69GB       47%
      Volume Guarantee                                     0B        0%
      Flexible Volume Metadata                        40.78MB        0%
      Deduplication Metadata                          29.20MB        0%
           Deduplication                              28.82MB        0%
           Cross Volume Deduplication                   384KB        0%
      Delayed Frees                                   105.6MB        0%
      File Operation Metadata                             4KB        0%
      Total Metadata Footprint                        297.0MB        0%
```

```
    Total Footprint                           5.74GB        6%
```

Additionally the `storage aggregate object-store show-space` will present a table of each aggregate and its utilization of cloud tier object storage.

While the amounts of data in this demo environment are relatively small, in a real environment a substantial portion of the SSD capacity can be reclaimed by tiering snapshots and cold data.

2.2.2 Changing a Volume’s Tiering Policy

As your organization’s needs change, you may need to adjust how FabricPool behaves for a given volume. In a previous exercise in this lab, you saw that you can change a volume’s tiering policy at the same time you attach an aggregate to the cloud tier, but now you will see how to change a volume’s tiering policy and cooling period independent of that process.

In this activity, you are tasked with altering a volume’s tiering policy from “Auto” to “Snapshot only” so that non-snapshot data no longer tiers to object storage.

1. You should still be on the Volume page for proj3 from the preceding exercise.

   In the volumes list, click **proj2**.

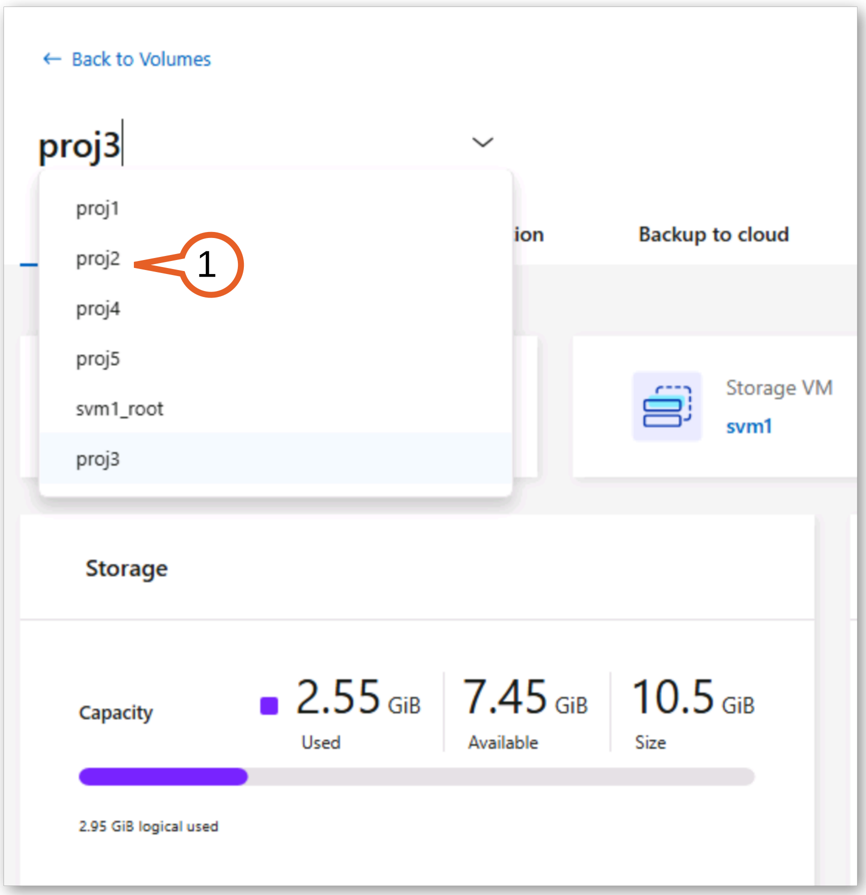

Figure 2-39:

2. On the proj2 volume page, scroll down to the Tiering section and notice that this volume currently has a tiering policy of “Auto” with approximately 2.7 GiB tiered to the cloud.

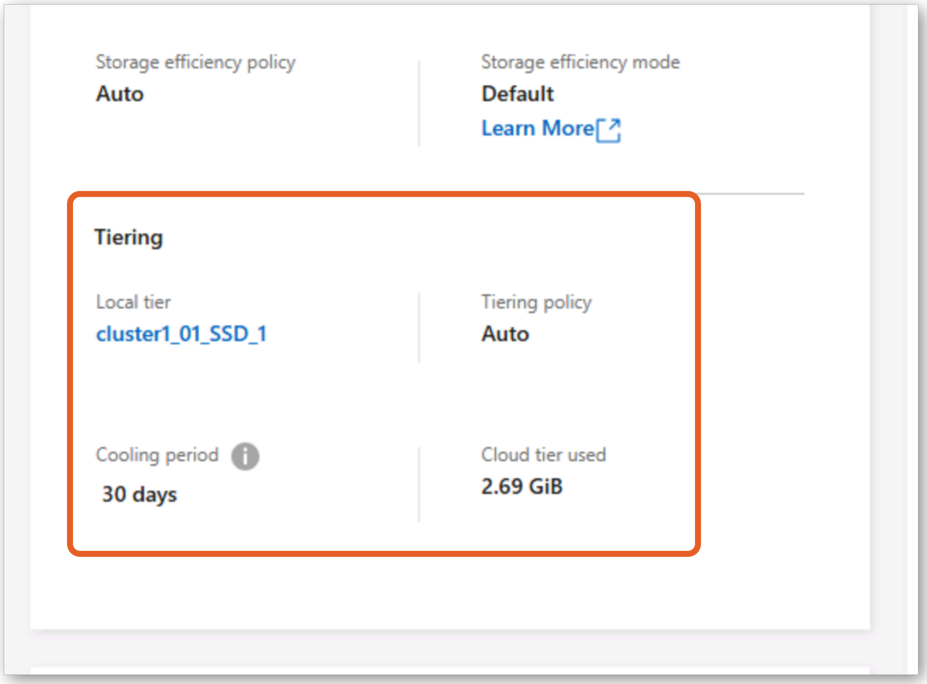

Figure 2-40:

3. Scroll back up and click the **More** menu in the top right.
4. Click **Edit Cloud Tiering Settings** from the drop-down menu.

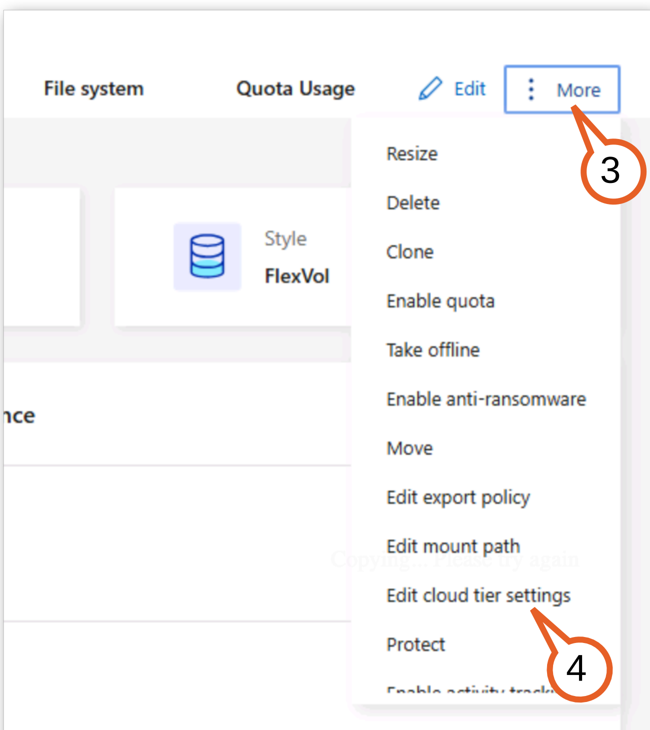

Figure 2-41:

5. From the Tiering Policy drop-down menu, select **Snapshot only**.
6. Beneath that is a field to set the Cooling Period, which governs how ONTAP determines what should be considered cold data.

   Change this value to **5** days.
7. Click **Save**.

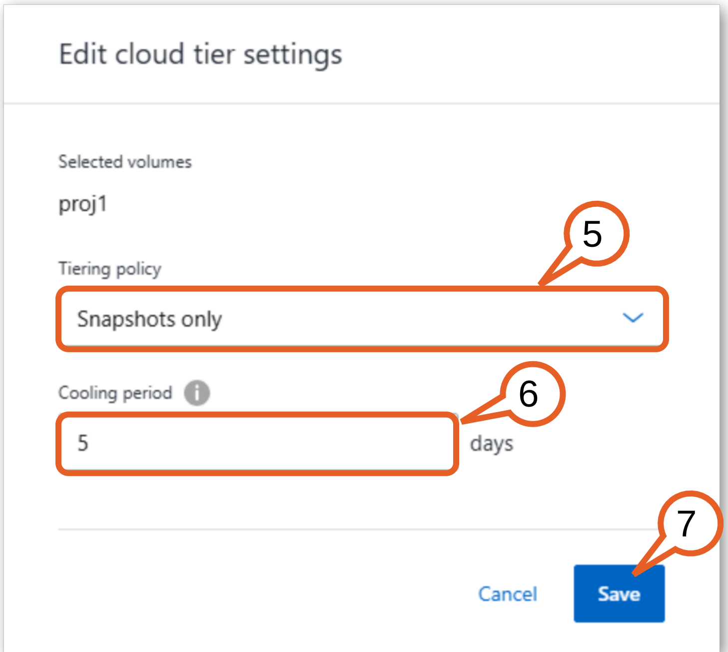

Figure 2-42:

8. The Tiering panel reports the new “Snapshot_only” value you just assigned, and the cooling period has changed to 5 days.
9. Notice that the existing data tiered to the cloud remains in the cloud tier for now. This data will be promoted back to the performance tier once it is accessed again. This is the default retrieval behavior. This behavior can be adjusted using Cloud Retrieval Policies which will be explored in a later activity.

   No additional data blocks on proj2 will tier out to the cloud tier until snapshot data is sufficiently cold to meet the cooling period you just set during a tiering scan, which occurs approximately every 24 hours unless manually triggered. If you had instead changed the tiering policy of the proj2 volume from “snapshot-only” to “auto”, snapshot blocks that were already out in proj2’s cloud tier would stay there until either the snapshots were deleted, or the snapshot blocks were accessed.

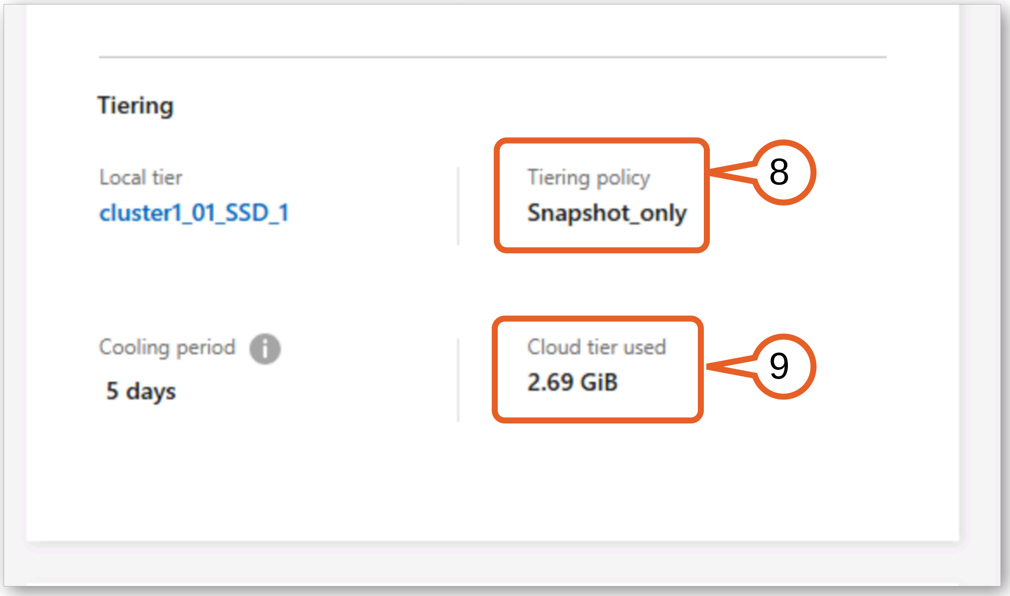

Figure 2-43:

Takeaway: FabricPool gives you the control and flexibility to adjust what data tiers and how quickly, down to tiering only older snapshots for a specific volume. These same tiering policies are available across Cloud Volumes ONTAP, FSx for ONTAP (where they govern data movement to the capacity pool tier), and Azure NetApp Files (where the coolness period controls when data moves to the cool tier).

2.2.3 Enable Cloud Write

Traditional FabricPool tiering is reactive: data must first be written to the local tier, remain untouched for the cooling period, and only then move to the cloud tier. That movement is carried out by a tiering scan, the background process that periodically scans the local tier to identify cold data blocks and move eligible blocks to the cloud tier. Cloud Write eliminates this bottleneck by allowing NFS clients to write data directly to the cloud tier, bypassing the local tier entirely. This is ideal for bulk data migrations, archival ingestion, and overflow scenarios where you want to preserve local tier capacity for hot workloads.

Note: Cloud Write supports NFS clients only. SMB and SAN clients continue writing to the local tier regardless of this setting.

Cloud Write has two prerequisites: the volume must use the **all** tiering policy, and the volume must be accessed through NFS (SMB clients always write to the local tier regardless of this setting).

1. From the Jumphost taskbar, click the **PuTTY** icon.

2. Double-click the **cluster1** saved session.

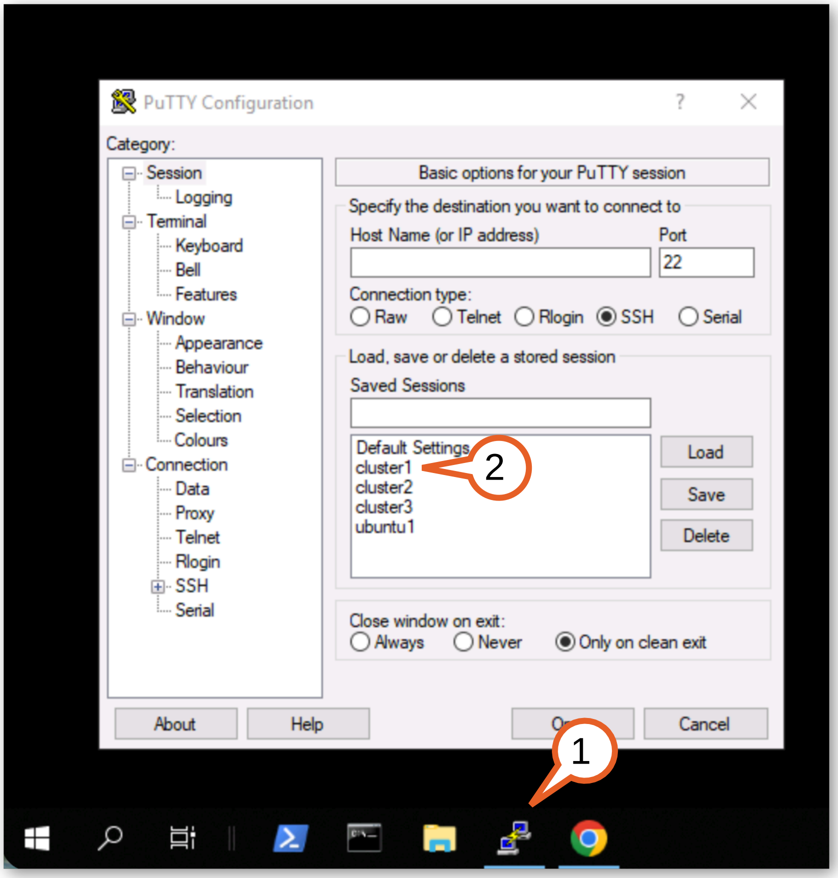

Figure 2-44:

3. When prompted, enter the password **`Netapp1!`**

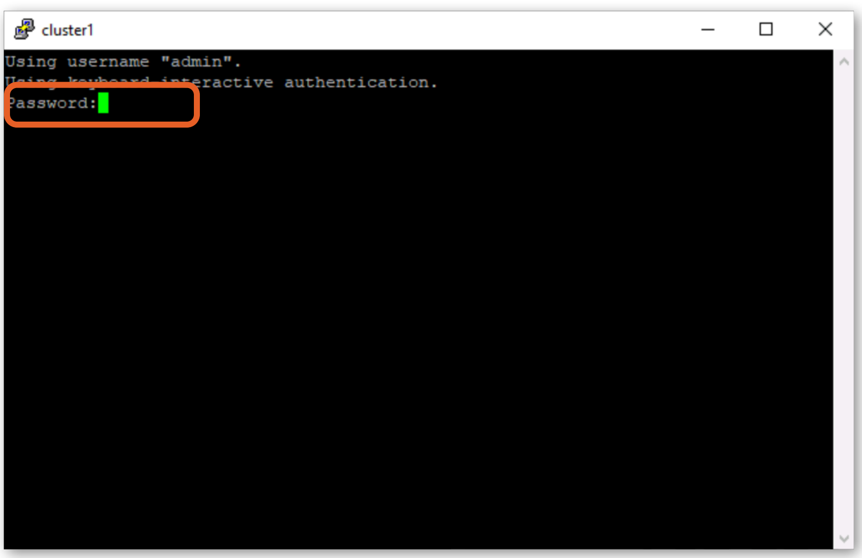

Figure 2-45:

4. In the PuTTY session, change to the advanced privilege level, entering `y` when prompted to confirm:

```
cluster1::> set -privilege advanced

Warning: These advanced commands are potentially dangerous; use them only when
         directed to do so by NetApp personnel.
Do you want to continue? {y|n}: y
```

5. Now, change the tiering policy on proj4 to "all" with the `volume modify` command:

```
cluster1::> volume modify -vserver svm1 -volume proj4 -tiering-policy all
Volume modify successful on volume proj4 of Vserver svm1.
```

   The "all" policy instructs ONTAP to tier all user data blocks immediately, without waiting for a cooling period. This is a prerequisite for Cloud Write. ONTAP will not allow direct cloud writes on a volume that might keep some data local based on access patterns.

6. Enable Cloud Write on the volume:

```
cluster1::*> volume modify -vserver svm1 -volume proj4 -is-cloud-write-enabled true
Volume modify successful on volume proj4 of Vserver svm1.
```

7. Confirm the setting is active:

```
cluster1::*> volume show -vserver svm1 -volume proj4 -fields is-cloud-write-enabled,tiering-
policy
vserver volume tiering-policy is-cloud-write-enabled
------- ------ -------------- ----------------------
svm1    proj4  all            true
```

   The output should show "is-cloud-write-enabled: true" and "tiering-policy: all".

8. From the Jumphost taskbar, right-click the **PuTTY** icon and select **PuTTY** from the menu.

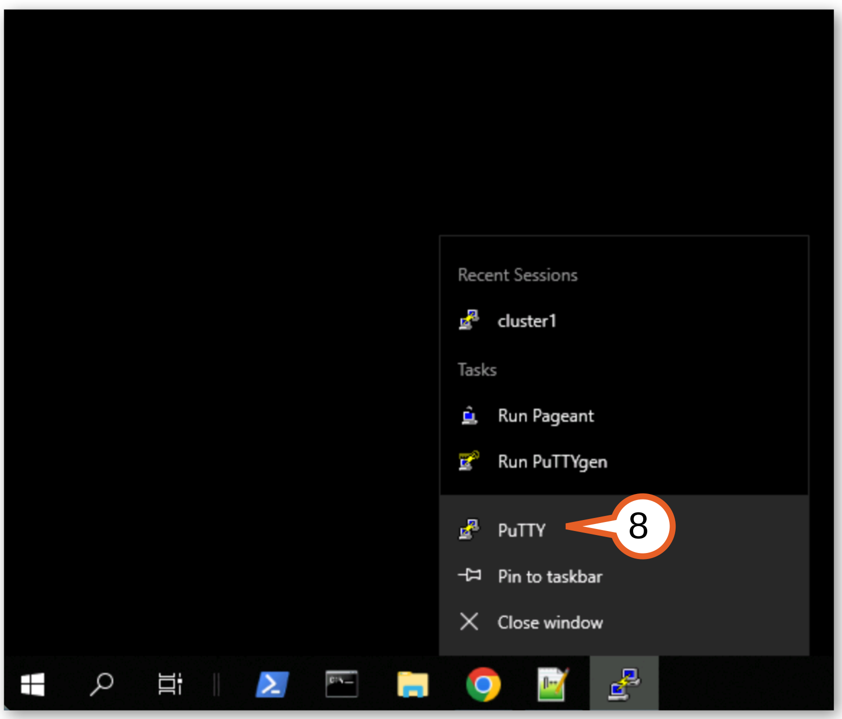

Figure 2-46:

9. Double-click the **ubuntu1** saved session.

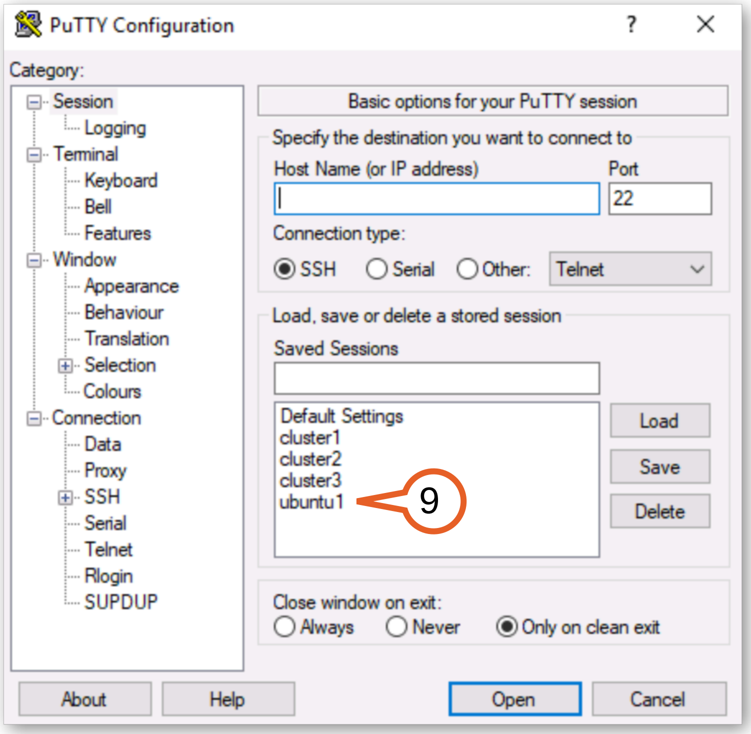

Figure 2-47:

10. When prompted, enter the password **`Netapp1!`**

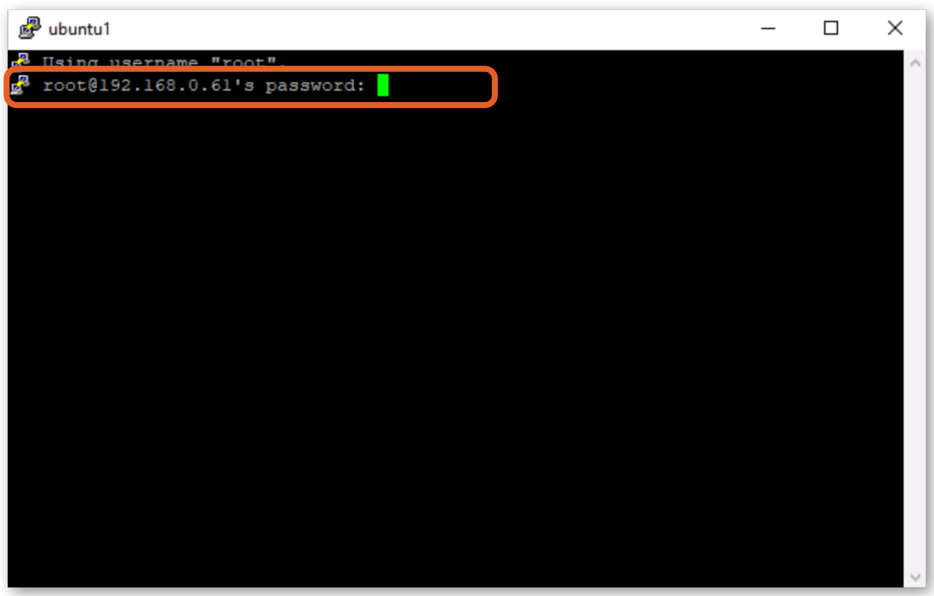

Figure 2-48:

11. Create a mount point for proj4:

```
root@ubuntu:~# mkdir -p /mnt/proj4
```

12. Mount the proj4 volume via NFS:

```
root@ubuntu:~# mount -t nfs 192.168.0.131:/proj4 /mnt/proj4
Created symlink /run/systemd/system/remote-fs.target.wants/rpc-statd.service # /lib/systemd/
system/rpc-statd.service.
```

13. Write a 1000 MB test file directly to the cloud tier:

```
root@ubuntu:~# dd if=/dev/zero of=/mnt/proj4/cloud_write_test bs=1M count=1000
1000+0 records in
1000+0 records out
1048576000 bytes (1.0 GB, 1000 MiB) copied, 4.35093 s, 241 MB/s
```

Because Cloud Write is enabled and the client is using NFS, the data flows directly to the cloud tier without consuming local tier capacity.

14. Now, switch back to the cluster1 PuTTY session, and verify the write went to the cloud tier by checking the footprint in both the Performance Tier (SSD) and the ontaps3_store:

```
cluster1::*> volume show-footprint -vserver svm1

...

      Vserver : svm1
      Volume  : proj4

      Feature                                          Used       Used%
      --------------------------------             ----------     -----
      Volume Data Footprint                            2.73GB        3%
             Footprint in Performance Tier            149.8MB        5%
             Footprint in ontaps3_store                2.73GB       95%
      Volume Guarantee                                     0B        0%
      Flexible Volume Metadata                        27.19MB        0%
      Deduplication Metadata                          64.54MB        0%
           Deduplication                              16.10MB        0%
           Cross Volume Deduplication                 48.44MB        0%
      Delayed Frees                                   96.73MB        0%
      File Operation Metadata                             4KB        0%
      Total Metadata Footprint                        244.8MB        0%

      Total Footprint                                  2.97GB        3%

      Footprint Data Reduction                        118.2MB        0%
           Data Compaction                            118.2MB        0%
      Effective Total Footprint                        2.86GB        3%
```

The used space on the local tier should not increase; instead you should see that the vast majority of the data (including the 1000MB file) resides directly in the cloud tier. A small amount of metadata is always written locally.

> **Takeaway:** Cloud Write fundamentally changes the data path for capacity-optimized workloads. Instead of filling the local tier and waiting for cooling, data lands directly on cheap object storage from the first write. This is particularly valuable for bulk migrations: terabytes of archive data can flow to the cloud tier without ever consuming SSD capacity or triggering tiering scans.

2.3 Object store operations

With data tiered to an external object store, you need the same operational confidence you have with local storage, including things like availability guarantees, lifecycle organization, and the ability to bring data back when priorities change. FabricPool provides these controls natively, giving administrators flexibility over how tiered data is protected, organized, and promoted back to the local tier without third-party tools or custom scripting.

Note: FabricPool mirroring and object tagging are available on on-premises ONTAP and Cloud Volumes ONTAP. These features are not currently exposed in FSx for ONTAP or Azure NetApp Files, where the cloud provider manages the object store infrastructure. Cloud retrieval behavior, however, is available across all ONTAP platforms.

In the following activities, you mirror a cloud tier for resiliency, tag tiered objects for lifecycle management, and retrieve data back to the local tier.

Activities
Add a FabricPool mirror
Apply object tags
Retrieve tiered data

2.3.1 Add a FabricPool mirror

FabricPool mirroring synchronously replicates tiered data to a second object store, giving you a safety net during object store maintenance, a migration path between providers, and disaster resilience for your cold data. Cluster 1 currently tiers to ONTAP S3 on Cluster 3. You will add StorageGRID as a mirror target so that both object stores hold an identical copy of all tiered data.

Mirroring is transparent to applications. ONTAP handles the synchronous writes to both targets automatically. If either object store becomes unavailable, ONTAP continues serving tiered data from the remaining target with no interruption.

1. Return to Cluster 1’s System Manager interface in the Chrome browser and navigate to **Storage** > **Tiers**.
2. Click **Add cloud tier** > **StorageGRID**.

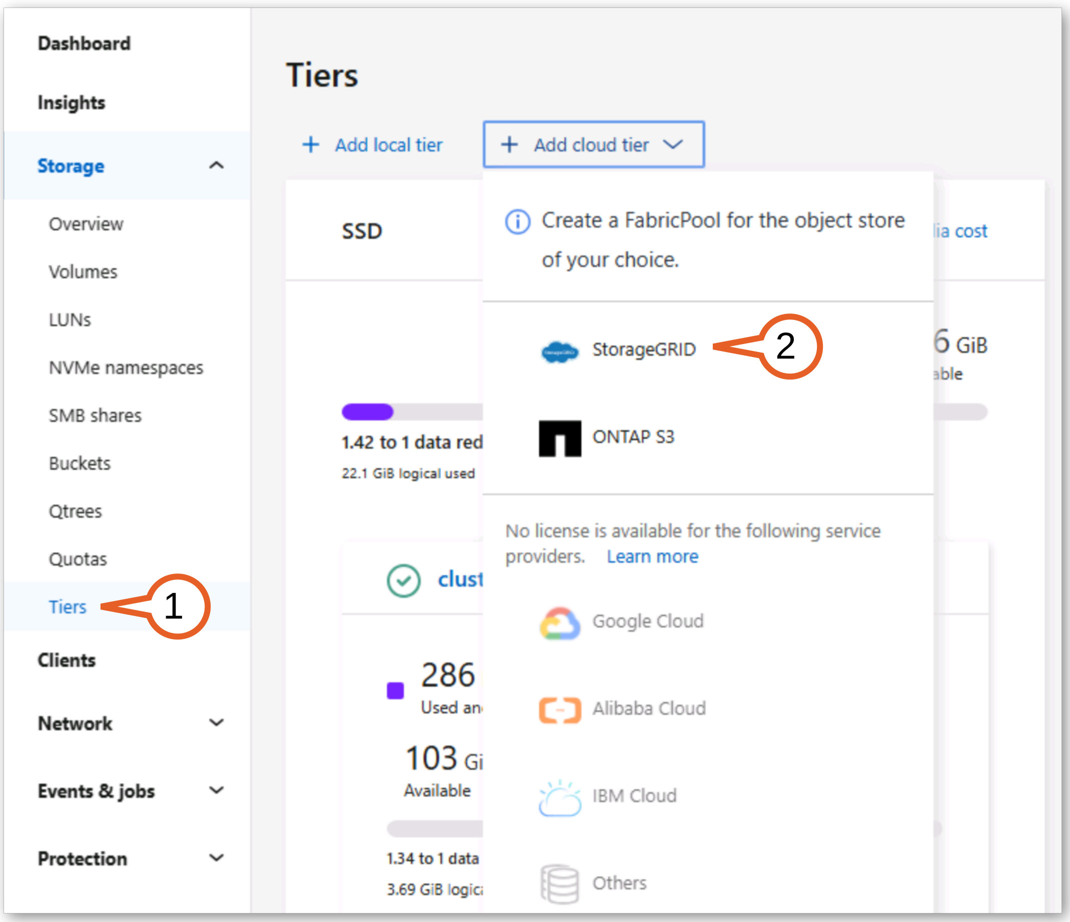

Figure 2-49:

3. Begin filling out the Add cloud tier form:

- Server Name (FQDN): **`dc1-adm1.demo.netapp.com`**
- Uncheck **SSL**.
- Port: **`10080`**

Note: SSL encryption is recommended for using FabricPool, but HTTP is used in this lab to simplify the setup process.

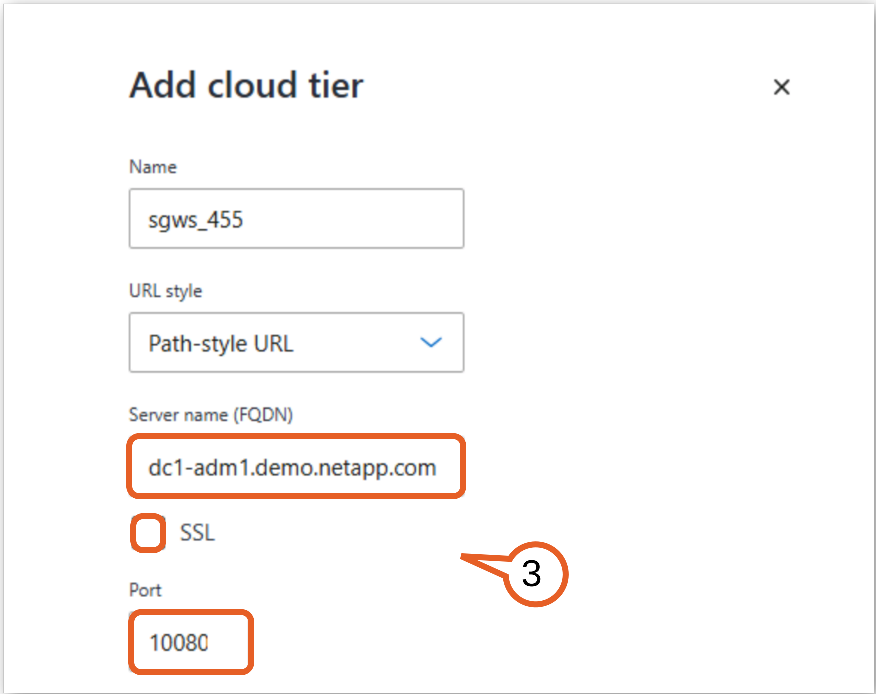

Figure 2-50:

4. Right-click on **Accounts** in the Chrome bookmarks bar and click **Open in new tab**.

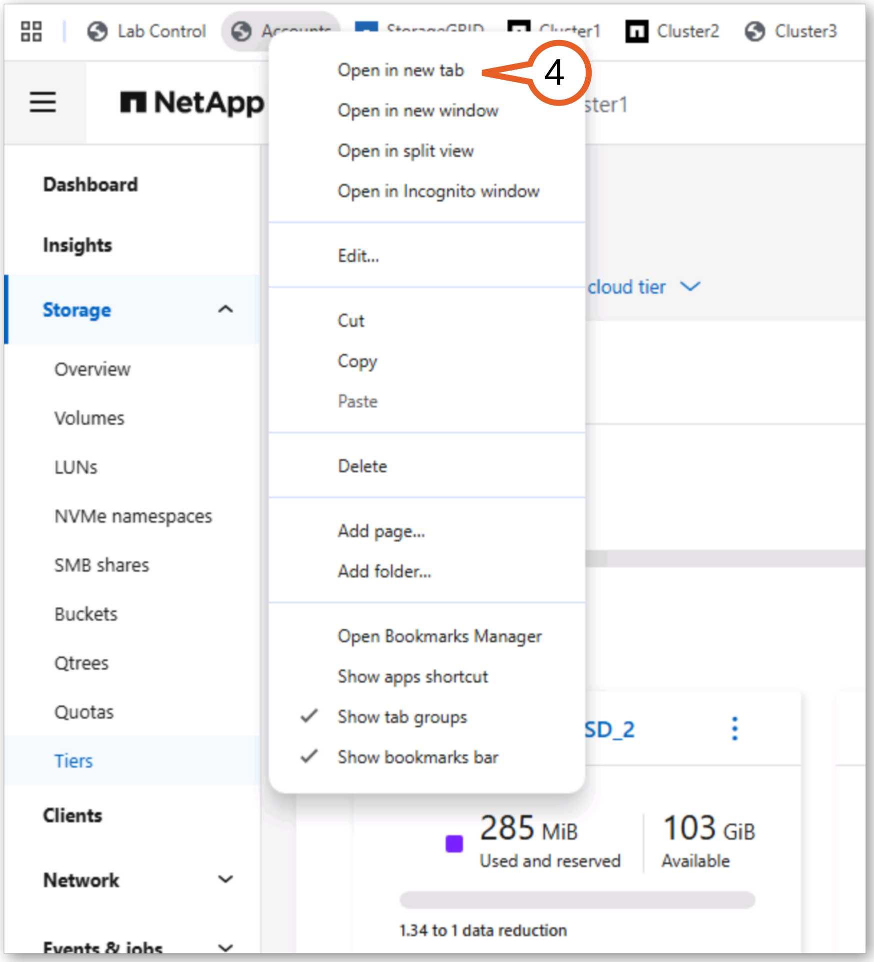

Figure 2-51:

5. On the Accounts page, locate the StorageGRID S3 access key and secret key listed under the StorageGRID credentials.

   Using the **copy buttons**, copy and then paste the Access key and Secret key credentials from the Accounts page into the "Add cloud tier" form:

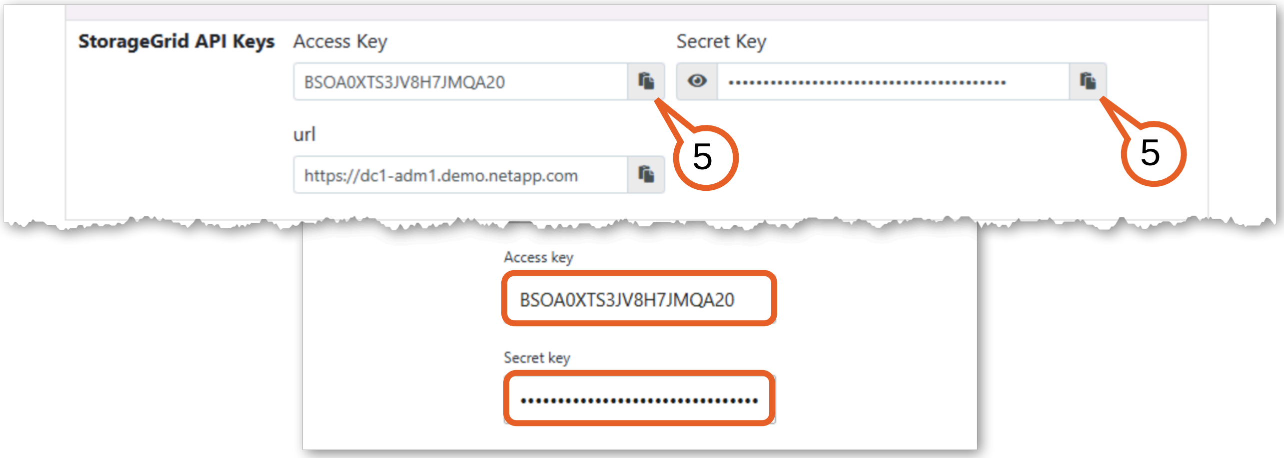

Figure 2-52:

6. Under Bucket name, enter **`cluster1-mirror`**.
7. Select **Create a new bucket**.

Figure 2-53:

8. Scroll down to the Add as FabricPool mirror section. You should see that both aggregates with an attached cloud tier are pre-selected.

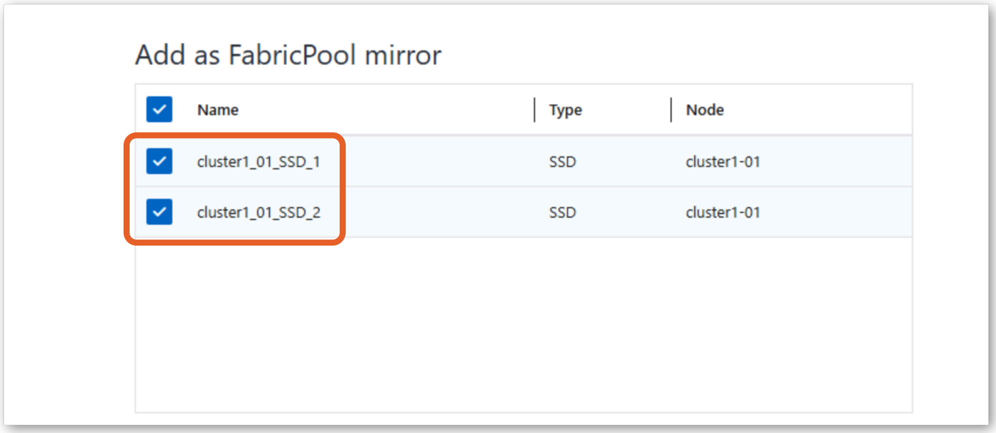

Figure 2-54:

9. Click **Save**.

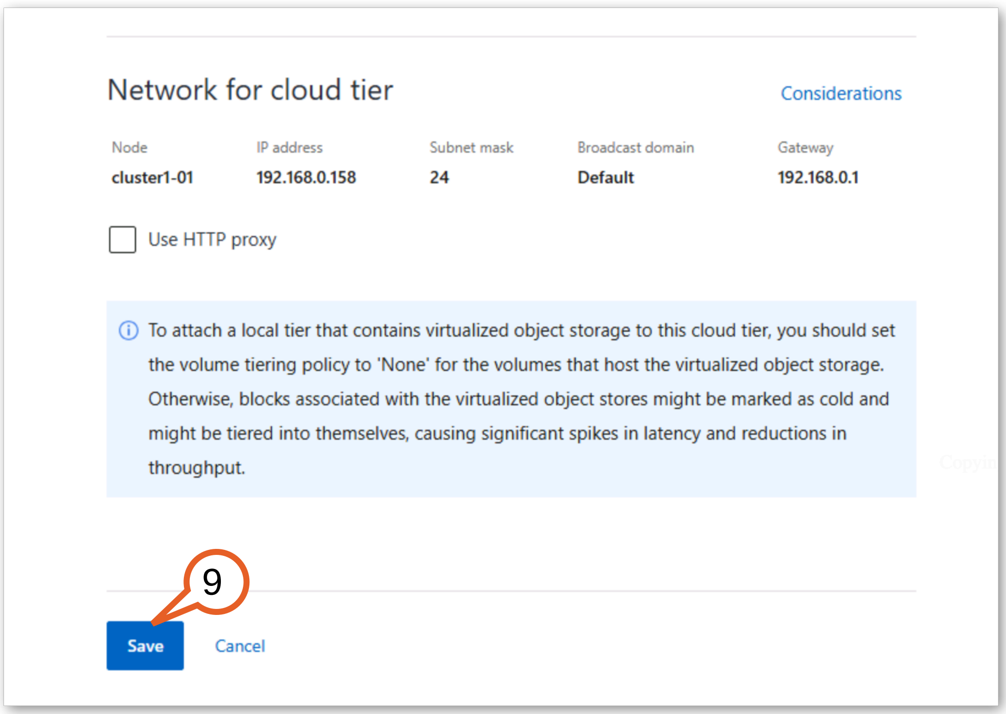

Figure 2-55:

10. You should see the new StorageGRID object store added to the Clouds section just beneath the OntapS3 section.

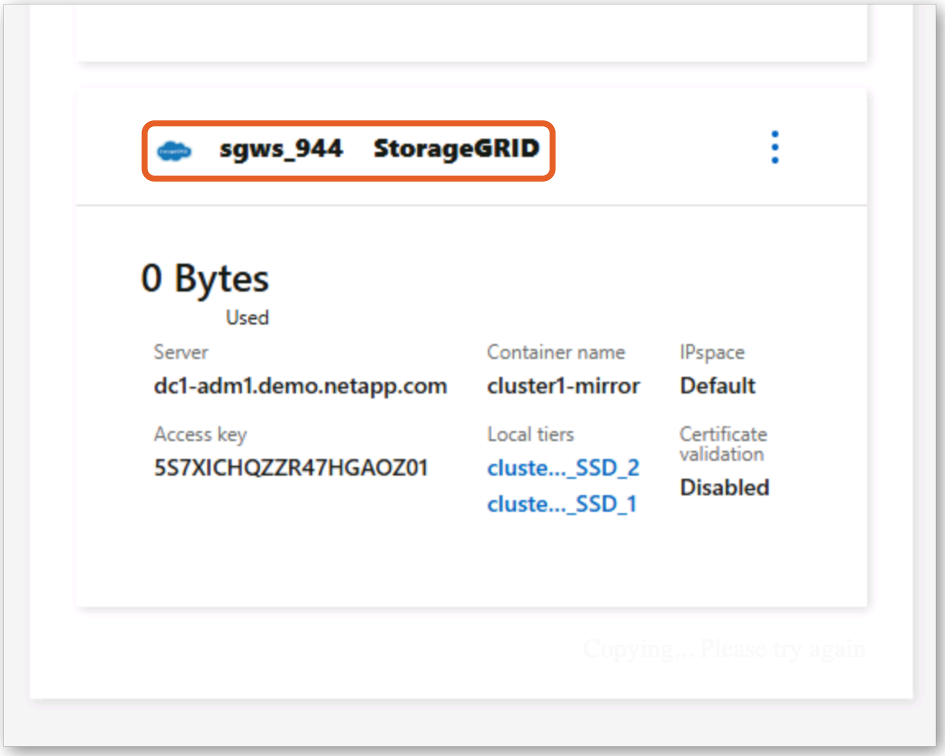

Figure 2-56:

11. Notice under the SSD section that the two mirrored aggregates now indicate that they are mirrored to the StorageGRID cloud tier with an icon on the bottom left of each panel. The data that has tiered to the ONTAP S3 cloud tier from these aggregates is currently syncing to the StorageGRID cloud tier.

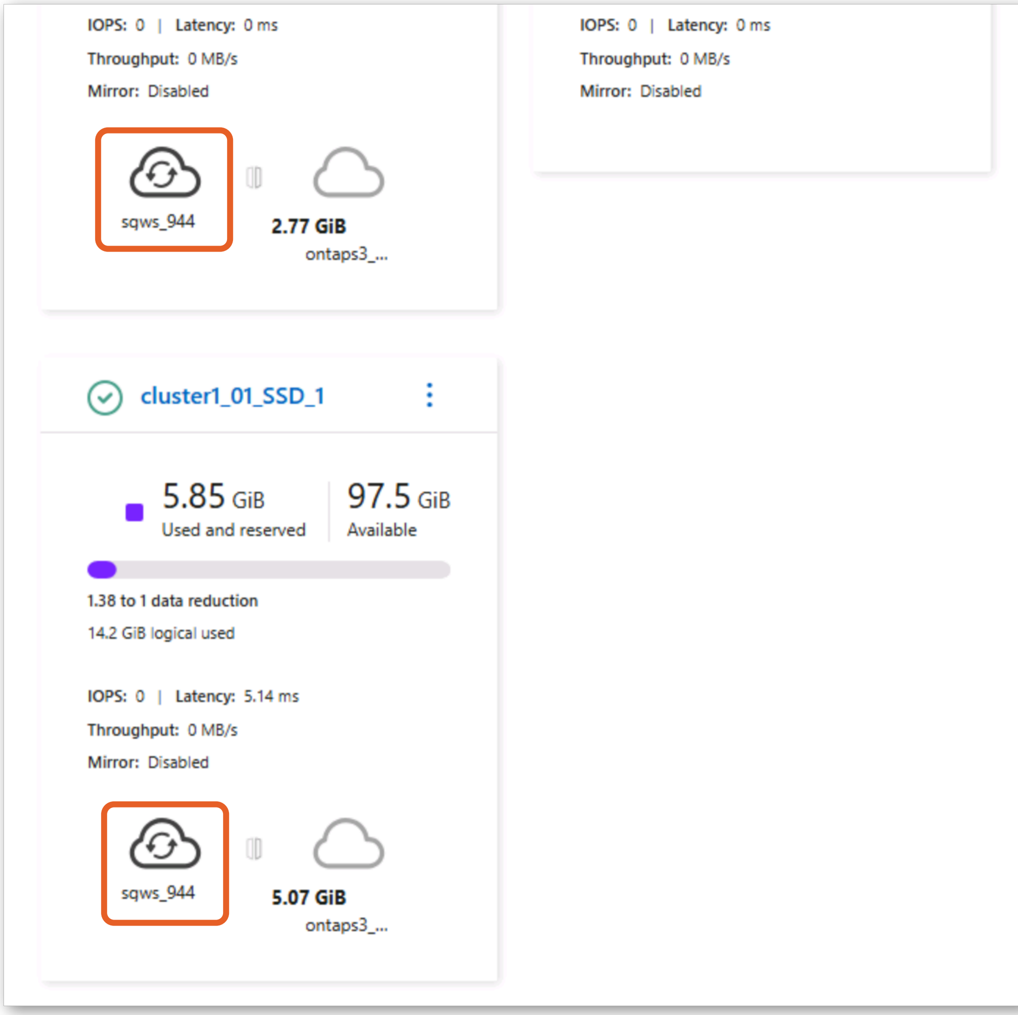

Figure 2-57:

The data that has tiered to the ONTAP S3 cloud tier from these aggregates is currently syncing to the StorageGRID cloud tier. After a few minutes these icons will show the amount of data mirrored. You can proceed to the next step without waiting.

Tip: When mirror synchronization is complete, ONTAP will allow you to swap between cloud tiers for a given aggregate as well as delete a mirror. This makes it easy to add or remove cloud tiers and migrate data between them. You can do this with the ONTAP CLI using the following command:

```
cluster1::> storage aggregate object-store modify -aggregate aggr1_cluster1 -object-store-name sgws_794 -mirror-type primary
```

You may try this in your lab instance once the mirror synchronization is complete, which can take up to 20 minutes.

Takeaway: NetApp is the only storage vendor that provides native dual-target tiering with automatic failover. Use FabricPool mirroring during planned maintenance windows (detach one target, maintain it, reattach), for migration between object store providers (mirror to the new target, then remove the old), or simply as an availability guarantee for your tiered data.

2.3.2 Apply object tags

FabricPool object tagging lets you attach key-value metadata to tiered objects based on the volume they come from. These tags are written to objects as they tier and can be used by the object store's lifecycle management engine, such as StorageGRID ILM rules, AWS S3 Lifecycle policies, or Azure Blob tiering rules, to apply different retention, replication, or storage class behaviors to different datasets without ONTAP involvement.

For example, you might tag volumes containing compliance data with `retention=7years` so that StorageGRID ILM can enforce immutability rules, while tagging general archive data with `tier=archive` for cost-optimized storage placement.

1. In the **cluster1** PuTTY session, apply an object tag to “proj1”:

   ```
   cluster1::*> volume modify -vserver svm1 -volume proj1 -tiering-object-tags
    "department=engineering"
   Volume modify successful on volume proj1 of Vserver svm1.
   ```

2. Apply a different tag to “proj3”:

   ```
   cluster1::*> volume modify -vserver svm1 -volume proj3 -tiering-object-tags "retention=7years"
   Volume modify successful on volume proj3 of Vserver svm1.
   ```

3. Verify the tags are assigned:

   ```
   cluster1::*> volume show -vserver svm1 -fields tiering-object-tags
   vserver volume tiering-object-tags
   ------- ------ ---------------------
   svm1    proj1  department=engineering
   svm1    proj2  -
   svm1    proj3  retention=7years
   svm1    proj4  -
   svm1    proj5  -
   svm1    svm1_root
                  -
   6 entries were displayed.
   ```

   The output shows each volume's assigned tags. Volumes without tags display an empty value.

4. To apply multiple tags to a single volume, separate them with commas:

   ```
   cluster1::*> volume modify -vserver svm1 -volume proj2 -tiering-object-tags
    department=marketing,priority=low
   Volume modify successful on volume proj2 of Vserver svm1.
   ```

5. Confirm the multi-tag assignment:

   ```
   cluster1::*> volume show -vserver svm1 -volume proj2 -fields tiering-object-tags
   vserver volume tiering-object-tags
   ------- ------ ---------------------------------
   svm1    proj2  department=marketing,priority=low
   ```

When you assign or modify tags on a volume, a background object tagging scanner automatically applies the updated tags to existing objects already in the cloud tier. You can check tagging progress with the vol show -fields needs-object-retagging command.

> **Takeaway:** Object tags bridge the gap between ONTAP's volume-level organization and the object store's bucket-level lifecycle policies. Instead of creating separate buckets for different data classes, you use tags to let StorageGRID ILM (or cloud lifecycle rules) automatically apply the correct retention, replication, or storage class, all within a single bucket.

2.3.3 Retrieve tiered data

There are situations where you need tiered data back on the local tier: before a performance-sensitive workload begins, when decommissioning a cloud tier, or when you want to guarantee local access speed for a specific dataset. ONTAP provides a cloud retrieval policy that controls whether tiered data is promoted back to the local tier on read, and a manual retrieval command for proactive bulk promotion.

In this activity, you take on a data repatriation scenario: you are decommissioning the cloud tier for the “proj1” volume and want to bring all of its tiered data permanently back on premises. This means you will both stop tiering the volume going forward and pull back the data that already resides in the cloud.

ONTAP supports three cloud retrieval policies:

- "default". The volume's tiering policy controls retrieval behavior. For example, with the "auto" policy, random reads promote data back to the local tier; with "all", no retrieval occurs. This is the default value for any volume.
- "on-read". All client-driven reads (both sequential and random) promote tiered data back to the local tier.
- "promote". Tiered data is proactively retrieved back to the local tier during the next tiering scan. The data promoted depends on the volume's tiering policy: with "none" all data returns; with "snapshot-only" only active file system data is retrieved. Use this before planned workload migrations or cloud tier decommissions.

Note: This activity continues in the “cluster1” PuTTY session from the previous activity, which should still be at the advanced privilege level. If you opened a new session, run **set -privilege advanced** (entering **y** to confirm) before continuing.

1. In the “cluster1” PuTTY session, check the current cloud retrieval policy on “proj1”:

```
cluster1::*> volume show -vserver svm1 -volume proj1 -fields cloud-retrieval-policy
vserver volume cloud-retrieval-policy
------- ------ ----------------------
svm1    proj1  default
```

   The default value is "default", meaning the volume's tiering policy governs retrieval. Since "proj1" uses the "auto" policy, random reads will promote data back, but sequential reads will not.

2. Because you are repatriating this volume, change its tiering policy to "none" so that ONTAP stops tiering new cold data to the cloud:

```
cluster1::*> volume modify -vserver svm1 -volume proj1 -tiering-policy none
Volume modify successful on volume proj1 of Vserver svm1.
```

   The "promote" cloud retrieval policy you set next is only supported with the "none" and "snapshot-only" tiering policies. Setting the policy to "none" both stops further tiering and makes all of the volume's cloud data eligible for bulk retrieval.

3. Change the retrieval policy to “promote” to trigger a bulk retrieval. Enter `y` when prompted to confirm.

```
cluster1::*> volume modify -vserver svm1 -volume proj1 -cloud-retrieval-policy promote

Warning: The "promote" cloud retrieve policy retrieves all of the cloud data for the specified
 volume. If the tiering policy is
         "snapshot-only" then only AFS data is retrieved. If the tiering policy is "none" then
 all data is retrieved. It may take a
         significant amount of time, and may degrade performance during that time. The cloud
 retrieve operation may also result in data
         charges by your object store provider.
Do you want to continue? {y|n}: y
Volume modify successful on volume proj1 of Vserver svm1.
```

   ONTAP will begin retrieving all tiered data for this volume back to the local tier during the next tiering scan cycle.

4. Confirm the policy changes:

```
cluster1::*> volume show -vserver svm1 -volume proj1 -fields tiering-policy,cloud-retrieval-
policy
vserver volume tiering-policy cloud-retrieval-policy
------- ------ -------------- ----------------------
svm1    proj1  none           promote
```

   The output should show “tiering-policy: none” and “cloud-retrieval-policy: promote”.

Takeaway: Cloud retrieval gives you an escape valve: the ability to proactively move data back to the local tier when performance requirements change. Combined with the on-read policy for workloads that gradually warm up, you have full control over data placement without needing to restructure volumes or change tiering policies permanently.

Tip: For FabricPool volumes with predominantly sequential read patterns (analytics, video editing, backup scans), ONTAP provides an Aggressive Read-Ahead option that prefetches all blocks in a file when any read occurs on that file. This keeps prefetched data cached locally so subsequent reads are served from the local tier rather than the cloud. Available on all platforms (including on-premises) beginning with ONTAP 9.14.1. Enable it per-volume at the advanced privilege level:

```
cluster1::*> volume modify -vserver svm1 -volume proj1 -aggressive-readahead-mode file_prefetch
Volume modify successful on volume proj1 of Vserver svm1.

cluster1::*> volume show -vserver svm1 -volume proj1 -fields aggressive-readahead-mode

vserver volume aggressive-readahead-mode
------- ------ -------------------------
svm1    proj1  file_prefetch
```

Set the value back to `none` to disable prefetching for random I/O workloads where prefetched blocks would go unused.

3 Summary

In this lab, you experienced the full FabricPool lifecycle, from initial setup through advanced capabilities and operational management. Here are the key takeaways:

- **Setup is straightforward**. FabricPool requires only an intercluster LIF, an S3 bucket, and a cloud tier configuration. The same workflow applies whether your target is StorageGRID, ONTAP S3, or a public cloud provider.
- **On-premises tiering is free**. There are no tiering license fees or capacity limits when using StorageGRID or ONTAP S3 as the object store target. You get cloud economics without cloud dependency.
- **Policies give you control**. Four tiering policies (auto, snapshot-only, all, none) let you customize behavior per volume. Different workloads on the same aggregate can each tier according to their own requirements.
- **Cloud Write eliminates bottlenecks**. For archival and migration workloads, data can bypass the local tier entirely. No more filling expensive SSDs with data that immediately needs to cool and tier.
- **Mirroring provides a safety net**. Dual-target tiering ensures your cold data remains available even during object store maintenance or migration.
- **You always retain control**. Object tags, retrieval policies, and enhanced space reporting give administrators full visibility and flexibility over where data lives and how it moves.
- **Skills transfer across the portfolio**. The FabricPool concepts you practiced here (tiering policies, cooling periods, retrieval, and monitoring) apply directly to Cloud Volumes ONTAP (AWS, Azure, GCP), Amazon FSx for NetApp ONTAP (capacity pool tier), and Azure NetApp Files (cool access). The underlying technology is the same; only the management interface and degree of automation differ.

FabricPool on ONTAP 9.18.1 transforms automatic tiering from a passive capacity optimization into an active data management platform, giving you the tools to write, read, protect, and retrieve tiered data with the same confidence you have for data on your local tier. Whether you manage an on-premises AFF cluster, run Cloud Volumes ONTAP in the public cloud, or consume FSx for ONTAP as a managed service, tiering ensures you pay for performance only where it matters, and let object storage handle the rest.

4 Additional information

This section provides reference material for the lab including documentation links, environment details, and version history.

4.1 Lab references

Use the following resources to learn more about FabricPool and the technologies covered in this lab:

- FabricPool overview: NetApp Documentation
- Set up StorageGRID as the cloud tier
- Enable or disable volume cloud write
- Create a FabricPool mirror
- FabricPool tiering policies
- Manage FabricPool mirrors
- TR-4598: FabricPool Best Practices
- Tier inactive Cloud Volumes ONTAP data to object storage
- FSx for ONTAP: Managing storage capacity and tiering
- Azure NetApp Files storage with cool access

4.2 Lab environment

This lab uses the following components:

Hostname	Description	IP Address(es)	Username	Password
Jumphost	Windows Server 2019 Remote Access host	192.168.0.5	DEMO\Administrator	Netapp1!
cluster1	NetApp ONTAP 9.18.1 cluster (pre-configured FabricPool tiering to ONTAP S3)	192.168.0.101	admin	Netapp1!
cluster2	NetApp ONTAP 9.18.1 cluster (clean for FabricPool setup)	192.168.0.102	admin	Netapp1!
cluster3	NetApp ONTAP 9.18.1 cluster (ONTAP S3 object store)	192.168.0.103	admin	Netapp1!
DC1	Active Directory Server	192.168.0.253	DEMO\Administrator	Netapp1!
dc1-adm1	StorageGRID Admin node	192.168.0.80	root	Netapp1!
ubuntu1	Ubuntu Linux NFS client	192.168.0.131	root	Netapp1!

All clusters run ONTAP 9.18.1. The intercluster network connects both ONTAP clusters to StorageGRID S3 endpoints. Production deployments should use HTTPS with CA-signed certificates; this lab uses HTTP to simplify setup.

4.3 Version history

Version	Date	Changes
1.0	May 2026	Initial release. Complete rewrite for ONTAP 9.18.1 covering FabricPool setup with StorageGRID, Cloud Write, Aggressive Read-Ahead, mirroring, object tagging, and cloud retrieval.

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  - Notice for ONTAP 9.10.1
  - Notice for ONTAP 9.9.1
  - Notice for ONTAP 9.8
  - Notice for ONTAP 9.7